



School of Computation, Information and Technology - Informatics
Technical University of Munich

Bachelor's Thesis in Informatics

An Empirical Comparison of Regret-Minimizing Algorithms

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Ein empirischer Vergleich von Regret-Minimierungsalgorithmen

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Abstract

This thesis empirically compares regret-minimizing algorithms under full-information and bandit feedback. All algorithms are evaluated within a common framework using expected action regret. The experiments cover two one-player environments—an adaptive historical-frequency process and an oblivious lazy random walk—as well as repeated Rock–Paper–Scissors and Rock–Paper–Scissors–Lizard–Spock games. Additional experiments examine how performance changes with the action-space size, horizon, and opponent.

The results show that similar performance under external regret can mask substantial differences in internal and swap regret. These differences are clearest in the adaptive environment and much smaller under the oblivious random walk. A related distinction appears in the repeated-game experiments. Several methods targeting internal or swap regret achieve smaller CE distances in the tested self-play settings, while external-regret learners can remain much closer to the CCE set than to the CE set. The cross-play results also show that finite-horizon regret can be sensitive to the opponent.

Overall, the results are broadly consistent with the qualitative expectations from the literature, while also showing that finite-horizon performance depends on the regret notion, feedback setting, environment, action-space size, horizon, and opponent.

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1 Introduction

1.1 Background and Motivation

Online learning studies sequential decision problems in which a learner repeatedly chooses among available actions without knowing future rewards or losses. A standard objective is to minimize *external regret*, which compares the learner with the best fixed action in hindsight. Classical work includes Hannan’s repeated-play framework [1], the Weighted Majority algorithm [2], and Hedge [3]. Under bandit feedback, where only the reward of the selected action is observed, algorithms such as EXP3 provide the corresponding adversarial guarantee [4].

External regret considers only fixed deviations. Internal regret allows one action to be replaced by another, while swap regret allows a separate replacement for every action. These stronger notions are particularly relevant in repeated games: vanishing external regret is associated with coarse correlated equilibrium [5], whereas vanishing internal or swap regret is associated with correlated equilibrium [6, 7]. This has motivated both direct regret-matching procedures and reductions that construct stronger-regret learners from external-regret algorithms [6–8].

Theoretical guarantees, however, do not by themselves determine finite-horizon behavior. Algorithms with the same asymptotic order can differ substantially over practical horizons, and stronger guarantees may come with different finite-horizon constants or update structures. The environment also matters: an adaptive reward process can react to the learner’s past actions, whereas an oblivious process is fixed independently of them. In repeated games, the opponent is itself part of the reward-generation process.

Structured self-play can introduce additional regularity beyond an arbitrary adversarial sequence. For example, optimistic methods can obtain faster regret rates in repeated games [5], and recent work shows that no-swap-regret dynamics can have stronger convergence properties in symmetric zero-sum self-play [9]. These results provide useful theoretical context, but they do not directly answer how the different algorithms compare empirically under a common finite-horizon evaluator.

This thesis therefore focuses on an empirical comparison. The goal is to study how external-, internal-, and swap-regret algorithms behave under the same experimental framework, how stable the observed differences are across feedback models and reward processes, and how they translate into repeated-game behavior.

1.2 Research Questions and Contributions

The empirical study is organized around three research questions:

1. **How do the considered algorithms differ in external, internal, and swap regret at finite horizons?**
2. **How stable are these differences across full-information and bandit feedback, adaptive and oblivious reward processes, action-space sizes, and horizons?**
3. **In repeated games, how are the regret differences reflected in CE/CCE behavior, and how strongly do they depend on the opponent?**

To address these questions, the thesis implements and evaluates a collection of existing external-regret, regret-matching, and swap-regret methods under a common framework. The experiments use expected action regret as the common regret metric. They cover full-information and bandit feedback, an adaptive historical-frequency environment, an oblivious lazy-random-walk environment, action-space and horizon scaling, and self-play and cross-play in RPS and RPSLS. The repeated-game experiments additionally evaluate the empirical joint-action distribution relative to the correlated-equilibrium and coarse-correlated-equilibrium sets.

The thesis does not propose a new learning algorithm or derive new regret bounds. Its contribution is the consistent empirical comparison of existing methods and the analysis of their finite-horizon behavior across these settings.

1.3 Thesis Structure

Chapter 2 introduces the online-learning setting, the regret notions used in the thesis, and the equilibrium concepts needed for the repeated-game experiments. Chapter 3 presents the implemented algorithms and the literature results most relevant to their interpretation. Chapter 4 describes the experimental environments, configuration, and evaluation metrics. Chapter 5 reports the empirical results. Chapter 6 discusses the main finite-horizon findings and relates them to the existing literature. Finally, Chapter 7 summarizes the conclusions and outlines directions for future work.

2 Theoretical Background

This chapter introduces the theoretical concepts underlying the implementation and empirical evaluation, including the online-learning setting, regret notions, equilibrium concepts, and their main connections.

2.1 Online Learning Setting

We consider a sequential decision problem with the finite action set $[K] = \{1, \dots, K\}$. At each round $t = 1, \dots, T$, the learner chooses an action without knowing the reward vector of that round in advance. Rather than selecting an action deterministically, the learner may randomize by choosing a probability distribution $p_t \in \Delta_K$, where

$$\Delta_K = \left\{ p \in \mathbb{R}_{\geq 0}^K : \sum_{i=1}^K p(i) = 1 \right\},$$

and then samples an action $I_t \sim p_t$.

Randomization prevents the learner from always following a predictable deterministic pattern and, under bandit feedback, also provides a mechanism for exploring different actions.

Associated with round t is a reward vector $r_t = (r_t(1), \dots, r_t(K)) \in [0, 1]^K$. After choosing I_t , the learner receives the reward $r_t(I_t)$. Its objective is to obtain high cumulative reward $\sum_{t=1}^T r_t(I_t)$.

The reward sequence may be generated independently of the learner's realized actions or may adapt to the previous interaction history. When rewards depend on past interaction, the environment is assumed to be *non-anticipating*: r_t may depend on previous actions, but not on the learner's fresh random draw I_t in the same round. The concrete reward environments used in the experiments are described in Chapter 4.

Feedback models. Under *full-information feedback*, the learner observes the complete reward vector r_t after each round.

Under *bandit feedback*, the learner observes only the reward $r_t(I_t)$ of the sampled action. The rewards of the remaining actions are hidden from the learner. EXP3 is a classical algorithm designed for this limited feedback setting [4].

The reward-generation process and the feedback model are separate aspects of an online-learning problem. The experiments in this thesis consider both full-information and bandit feedback.

Throughout the thesis, algorithms are described in terms of rewards. Results stated in terms of losses can be translated using $\ell_t(i) = 1 - r_t(i)$.

2.2 Regret Notions

Absolute cumulative reward alone is often insufficient to evaluate an online learner, because the attainable reward depends strongly on the environment. Regret instead compares the learner's decisions with alternative decisions that could have been applied to the same reward sequence.

Throughout this thesis, regret computed from the actions actually sampled by the learner is called *action regret*, following the terminology of Greenwald et al. [10]. This is distinct from *distribution regret*, which is computed from the learner's mixed strategy rather than from the sampled actions. For two actions $i, j \in [K]$, define the cumulative replacement gain of the sampled action sequence by

$$G_T(i, j) = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)).$$

This quantity measures how much additional reward would have been obtained if every realized occurrence of action i had instead been replaced by action j .

The matrix G_T is the common basis for all three regret notions used in the experiments.

Definition 2.1 (External regret) *The external regret after T rounds is*

$$R_T^{\text{ext}} = \max_{j \in [K]} \sum_{t=1}^T (r_t(j) - r_t(I_t)).$$

Equivalently,

$$R_T^{\text{ext}} = \max_{j \in [K]} \sum_{i=1}^K G_T(i, j).$$

External regret compares the realized trajectory with the best single fixed action in hindsight.

Definition 2.2 (Internal regret) *The internal regret after T rounds is*

$$R_T^{\text{int}} = \max_{i, j \in [K]} G_T(i, j).$$

Internal regret asks whether replacing one particular action i by another action j would have systematically improved the learner's realized performance [6].

Definition 2.3 (Swap regret) *A swap deviation is a mapping $\phi : [K] \rightarrow [K]$ that assigns a replacement action to each source action. The swap regret is*

$$R_T^{\text{swap}} = \max_{\phi: [K] \rightarrow [K]} \sum_{t=1}^T (r_t(\phi(I_t)) - r_t(I_t)).$$

Because the replacement for each source action can be chosen independently, this is equivalently

$$R_T^{\text{swap}} = \sum_{i=1}^K \max_{j \in [K]} G_T(i, j).$$

Swap regret therefore allows a separate best replacement for every action selected by the learner [7, 11]. The external-, internal-, and swap-regret quantities above are all action-regret quantities because they are evaluated on the sampled action sequence.

Expected action regret. For a randomized learner, action regret is itself a random variable. For $q \in \{\text{ext}, \text{int}, \text{swap}\}$, this thesis uses

$$\mathcal{R}_T^q = \mathbb{E}[R_T^q]$$

for the corresponding *expected action regret*. The expectation is over the randomness of the interaction: the learner’s randomization and, where applicable, random environments or other randomized players. In the swap case, this convention agrees with the “expected swap regret” studied by Huang and Pan [12], where the best swap deviation is selected inside the expectation.

Huang and Pan distinguish this quantity from swap pseudo-regret, where the maximization over deviations is taken outside the expectation [12]. The two notions are therefore not treated as interchangeable in this thesis. The experiments estimate expected action regret by first computing action regret separately for each replicate and then averaging the resulting values.

Relations between the regret notions. The three notions differ in the deviation class used for comparison. External regret considers one fixed replacement, internal regret one source–target replacement, and swap regret an arbitrary replacement mapping. Blum and Mansour note that external and internal regret are both upper-bounded by swap regret, while swap regret is at most K times internal regret; external and internal regret are not generally ordered [7]. Thus swap regret is the strongest of the three benchmarks considered here, while external and internal regret capture different kinds of deviations.

The corresponding deviation classes are summarized in Table 2.1.

Table 2.1: Deviation classes associated with the regret notions.

Regret notion	Hindsight deviation
External	Replace all realized actions by one fixed action
Internal	Replace one action i by another action j
Swap	Choose a replacement separately for every action

For a single realized action sequence, the experiments compute cumulative action regret R_T and average action regret R_T/T . A learner is no-regret with respect to a given deviation class if

$$\limsup_{T \rightarrow \infty} \frac{R_T}{T} \leq 0.$$

For internal and swap regret as defined above, the identity replacement makes the regret nonnegative, so this reduces to $R_T/T \rightarrow 0$.

For comparison, replacing $\mathbf{1}\{I_t = i\}$ in $G_T(i, j)$ by $p_t(i)$ gives

$$\overline{G}_T(i, j) = \sum_{t=1}^T p_t(i) (r_t(j) - r_t(i)).$$

This is the corresponding *distribution-regret* construction in the terminology of Greenwald et al. [10]: it evaluates the mixed strategy rather than the sampled action sequence. Theoretical guarantees therefore need not refer to the same regret object or probabilistic criterion: the literature distinguishes action from distribution regret [10] and pseudo-regret from stronger high-probability guarantees [12]. These distinctions are stated explicitly where the algorithms are introduced in Chapter 3.

The experiments in this thesis use action regret under both full-information and bandit feedback, and report its replicate mean as an empirical estimate of expected action regret. In the bandit experiments, the learner still receives only $r_t(I_t)$; rewards of actions that were not selected are retained by the experimental environment solely for offline regret evaluation.

2.3 Equilibrium Concepts

The repeated-game experiments require several standard equilibrium concepts. A finite normal-form game consists of a set of players $N = \{1, \dots, n\}$, finite action sets A_i , and payoff functions $u_i : A_1 \times \dots \times A_n \rightarrow \mathbb{R}$.

Let $a_t = (a_{1,t}, \dots, a_{n,t})$ be the realized joint action in round t . The empirical distribution of joint play after T rounds is

$$\hat{\mu}_T(a) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a_t = a\}.$$

This distribution is used directly in the equilibrium analysis of the experiments.

Definition 2.4 (Coarse correlated equilibrium) A distribution μ over joint actions is a coarse correlated equilibrium (CCE) if, for every player i and every fixed alternative action $a'_i \in A_i$,

$$\sum_a \mu(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq 0.$$

A CCE therefore rules out profitable deviations to a single fixed action chosen independently of the action drawn from μ . Replacing the right-hand side 0 by $\varepsilon \geq 0$ gives the corresponding ε -CCE notion [9].

Definition 2.5 (Correlated equilibrium) A distribution μ over joint actions is a correlated equilibrium (CE) if, for every player i and every pair of actions $a_i, a'_i \in A_i$,

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq 0.$$

Unlike CCE, a correlated equilibrium rules out deviations that depend on the action assigned to the player. Replacing the right-hand side 0 by $\varepsilon \geq 0$ defines an ε -CE. This parallels the action-dependent replacement considered by internal and swap regret. Correlated equilibrium was introduced by Aumann [13].

Every correlated equilibrium is also a coarse correlated equilibrium, so $\text{CE}(G) \subseteq \text{CCE}(G)$ [9].

The experiments later measure the distance of $\hat{\mu}_T$ to these equilibrium sets; the precise numerical procedure is described in Chapter 4.

2.4 Regret and Equilibrium

The standard regret–equilibrium correspondence motivates the repeated-game experiments. If every player has vanishing external regret, the empirical distribution of joint play approaches the coarse correlated-equilibrium set [5]. If every player has vanishing internal or swap regret, the corresponding empirical distribution approaches the correlated-equilibrium set [6, 7]. These standard implications are used later only as qualitative reference points for the empirical CE and CCE measurements.

A stronger result is known in a more specialized self-play setting. Paes Leme et al. [9] show that, for almost all symmetric two-player zero-sum games, if both players use the same no-swap-regret dynamics with symmetric initialization, then all but a vanishing fraction of the mixed-strategy iterates are close to a symmetric Nash equilibrium. The experiments in this thesis do not test this frequent-iterate result directly; they measure regret and the time-averaged joint distribution’s distance to CE and CCE.

The next chapter introduces the concrete learning algorithms implemented in the empirical framework and summarizes their theoretical regret guarantees.

3 Regret-Minimizing Algorithms

This chapter introduces the algorithms used in the experiments. All methods are taken from the literature; the purpose here is only to state the implemented variant, the main idea behind it, and the theoretical guarantee that is most relevant for the later empirical comparison. Detailed proofs are not repeated.

The implementations fall into three groups: standalone external-regret learners, regret-matching procedures, and reductions that construct stronger regret guarantees from external-regret sublearners.

3.1 External-Regret Learners

Hedge and Optimistic Hedge. Hedge is the main full-information external-regret baseline. The implementation samples according to exponential weights and, for a known horizon, uses $\eta = \sqrt{\frac{8 \log K}{T}}$ [3]. With bounded rewards this gives the standard root- T distribution-regret guarantee. The same fixed-horizon Hedge learner is used inside BM-Hedge. Ito-Hedge instead uses the anytime version $\eta_s = \sqrt{8 \log K / s}$, where s is the local update count of the selected Ito sublearner.

Following Chen and Peng [5], Optimistic Hedge (OptHedge) uses the optimistic update

$$p_{t+1}(i) \propto p_t(i) \exp(\eta(2r_t(i) - r_{t-1}(i)))$$

with learning rate $\eta = \left(\frac{\log K}{T}\right)^{1/6}$. Their fast $O(T^{1/6} \log^{5/6} K)$ mixed external-regret result applies when both players use Optimistic Hedge in a two-player repeated game. It is not a general adversarial guarantee, so the one-player OptHedge experiments are used only as empirical comparisons.

EXP3, EXP3-IX, and Tsallis-INF. EXP3 is the explicit-exploration bandit baseline. Following Auer et al. [4], the standalone implementation uses

$$\gamma = \min \left\{ 1, \sqrt{\frac{K \log K}{(e-1)T}} \right\}, \quad \eta = \frac{\gamma}{K},$$

together with the usual importance-weighted reward estimate. The EXP3 learners inside BM-EXP3 use the same update rule but the Blum–Mansour tuning $\gamma = \min\{1, \sqrt{K \log K / T}\}$ [7].

EXP3-IX replaces explicit uniform exploration by implicit exploration. In loss form, the estimator divides the observed loss by $p_t(i) + \gamma$, and no additional uniform mixture is used. Following Neu [14], the implementation uses

$$\eta = \sqrt{\frac{2 \log K}{KT}}, \quad \gamma = \frac{\eta}{2}.$$

Neu proves a root- T -order high-probability realized external-regret bound.

Tsallis-INF is used only as the inner bandit learner of Ito-Tsallis. Following Zimmert and Seldin [15], the implementation uses the importance-weighted $\alpha = 1/2$ variant with local learning rate $\eta_s = 1/\sqrt{s}$. This is an equivalent scaling of their formulation, which gives adversarial pseudo-regret of root- T order.

3.2 Regret Matching

Both regret-matching implementations maintain cumulative pairwise replacement gains

$$C_t(i, j) = \sum_{s=1}^t \mathbf{1}\{I_s = i\} (r_s(j) - r_s(i)).$$

With rewards in $[0, 1]$, the implementation sets the fixed normalization to K , which is a valid choice for the Hart–Mas-Colell procedure [6], and forms a stochastic transition matrix using

$$Q_t(i, j) = \frac{[C_t(i, j)]_+}{Kt}, \quad i \neq j,$$

with diagonal entries chosen so that each row sums to one.

Regret Matching (RM) implements the adaptive procedure of Section 2 of Hart and Mas-Colell [6]: after playing action i , the next distribution is the i -th row of Q_t . Hart and Mas-Colell prove that if all players use this procedure, the empirical distribution of play converges almost surely to the correlated-equilibrium set. This is a joint self-play result, not a general individual no-internal-regret guarantee against arbitrary adaptive opponents.

Stationary Regret Matching (SRM) instead plays an invariant distribution of the transition matrix,

$$p_{t+1} = p_{t+1} Q_t.$$

Theorem A of Hart and Mas-Colell [6] gives vanishing individual pairwise regret almost surely under the paper’s conditional-independence assumption. This distinction between RM and SRM is important in the later one-player and cross-play comparisons.

3.3 Swap-Regret Reductions

Blum–Mansour. The Blum–Mansour reduction maintains K external-regret learners, one for each source action [7]. Their output distributions form a stochastic matrix Q_t , and the master distribution is chosen stationary:

$$p_t = p_t Q_t.$$

Each row can be interpreted as learning a replacement distribution for one source action.

In the full-information version, all rows are updated every round with the weighted reward vector $p_t(i)r_t$. BM-Hedge uses the fixed-horizon Hedge learner above. BM-OptHedge replaces Hedge by Optimistic Hedge and, following Chen and Peng [5], uses

$$\eta = \left(\frac{K \log K}{m^2 T} \right)^{1/4}.$$

In repeated games, m is the actual number of players; the one-player benchmark keeps the implementation default $m = 2$. The fast $O((K \log K)^{3/4} m^{1/2} T^{1/4})$ mixed swap-regret guarantee of Chen and Peng [5] applies when all players use this optimistic self-play dynamic.

BM-EXP3 is the bandit Blum–Mansour variant. It uses EXP3 row learners and updates all K rows every round using the importance-weighted feedback specified by Blum and Mansour [7]. The corresponding guarantee is on swap pseudo-regret.

Ito reduction. Ito retains the stationary-distribution construction but updates only one row per round [8]. First

$$J_t \sim p_t,$$

then the played action is sampled from row J_t , and only that selected sublearner receives feedback. Because $p_t = p_t Q_t$, the marginal distribution of the played action is still p_t .

Ito-Hedge uses the anytime Hedge learner, while Ito-Tsallis uses importance-weighted Tsallis-INF. Their learning rates depend on the selected row’s local update count rather than on the global round. Ito’s generic reduction shows that an external learner with expected regret $r(S)$ yields $Kr(T/K)$ for the maximization-outside-expectation swap criterion [8]. In the terminology of Section 2.2, this is swap pseudo-regret.

LCE-IX. LCE-IX is the bandit swap-regret method of Huang and Pan [12]. Like Blum–Mansour it uses a stationary distribution and updates all rows every round, but combines this construction with implicit exploration. Following the parameter schedule in the same work, the implementation uses

$$\eta_t = \sqrt{\frac{\log K}{t}}, \quad \gamma_t = \frac{\eta_t}{2}.$$

Huang and Pan prove $O(K\sqrt{T \log K})$ expected swap regret—an expected action-regret quantity in the terminology of Chapter 2—and a corresponding high-probability bound for instantaneous swap regret.

3.4 Algorithm Summary

Table 3.1 summarizes the role of each implementation and the literature guarantee most closely associated with the implemented parameterization. All displayed big- O rates are cumulative in the horizon T and suppress constant factors; the RM and SRM entries instead

state their asymptotic average-regret guarantees. They are not directly comparable unless the regret object and the assumptions indicated below the table agree.

Table 3.1: Algorithms used in the experiments and their principal literature guarantees.

Algorithm	Feedback	Main role	Regret notion	Literature guarantee
Hedge	Full	External baseline	External distribution regret	$O(\sqrt{T \log K})$
OptHedge*	Full	Optimistic external regret	External distribution regret	$O(T^{1/6} \log^{5/6} K)$
RM**	Full	Adaptive regret matching	Internal action regret	$R_T^{\text{int}}/T \rightarrow 0$ a.s.
SRM**	Full	Stationary pairwise regret	Internal action regret	$R_T^{\text{int}}/T \rightarrow 0$ a.s.
BM-Hedge	Full	BM swap reduction	Swap distribution regret	$O(\sqrt{TK \log K})$
BM-OptHedge*	Full	Optimistic BM reduction	Swap distribution regret	$O(m^{1/2} (K \log K)^{3/4} T^{1/4})$
Ito-Hedge	Full	One-row Ito reduction	Swap pseudo-regret	$O(\sqrt{TK \log K})$
EXP3***	Bandit	External baseline	External pseudo-regret	$O(\sqrt{TK \log K})$
EXP3-IX****	Bandit	Implicit-exploration baseline	Realized external action regret	$O(\sqrt{TK \log K})$ w.h.p.
BM-EXP3	Bandit	BM swap reduction	Swap pseudo-regret	$O(K \sqrt{KT \log K})$
Ito-Tsallis	Bandit	One-row Ito reduction	Swap pseudo-regret	$O(K \sqrt{T})$
LCE-IX	Bandit	Stationary swap regret	Expected swap action regret	$O(K \sqrt{T \log K})$

*The optimistic fast rates are repeated-game results rather than generic adversarial guarantees: the OptHedge bound assumes two-player self-play with both players using Optimistic Hedge, while the BM-OptHedge bound assumes all m players use the optimistic BM dynamic [5].

**For RM, the Hart–Mas-Colell Main Theorem assumes that all players use the adaptive procedure. For SRM, the individual vanishing-regret theorem allows arbitrary opponent behavior subject to conditional independence of players’ current randomizations given the past [6].

***The displayed EXP3 rate is the pseudo-regret guarantee for the nonstochastic adversarial-bandit model of Auer et al. [4].

****For EXP3-IX the displayed order suppresses the confidence-dependent term in Neu [14]; the theorem is a high-probability bound on realized external regret.

The central implementation differences for the empirical comparison are therefore simple: standalone learners target external regret; RM and SRM react directly to pairwise replacement regret; BM and LCE-IX update all row learners through a stationary construction; and Ito updates only one selected row per round. The next chapter describes how these methods are evaluated under a common experimental framework.

4 Experimental Methodology

This chapter describes the experimental setup used in Chapter 5. The experiments combine two one-player reward processes with repeated play in RPS and RPSLS. Full-information and bandit learners are evaluated with the same expected action-regret metrics introduced in Chapter 2. Repeated-game experiments additionally track the empirical joint-action distribution and its distance to the CE and CCE sets.

4.1 Experimental Setup

The main trajectory experiments use horizon $T = 10^6$, 12 independent replicates, and base seed 42. All random-number streams are deterministically derived from this base seed, so rerunning the same experiment configuration reproduces the same stochastic trajectories. The one-player experiments use $K = 10$ actions unless the action-space size is varied. The full-information and bandit algorithm sets are those summarized in Chapter 3.

The additional scaling experiments are:

- **Action-space scaling:** HFA with $K \in \{2, 3, 5, 10, 20, 30, 50\}$ and $T = 10^6$.
- **One-player horizon scaling:** HFA and LRW with $K = 10$.
- **Repeated-game horizon scaling:** RPSLS self-play and selected cross-play profiles.

The horizon-scaling experiments use

$$\{1000, 2000, 3000, 5000, 10^4, 2 \cdot 10^4, 3 \cdot 10^4, 5 \cdot 10^4, 10^5, 2 \cdot 10^5, 3 \cdot 10^5, 5 \cdot 10^5, 10^6\}.$$

Each horizon is run as an independent experiment: the learner is reinitialized, and any fixed-horizon parameter is computed from that run's own T .

4.2 One-Player Environments

The two one-player environments are deliberately different in how the reward sequence is generated.

4.2.1 Adaptive Historical-Frequency Environment

The historical-frequency adversary (HFA) is *adaptive but non-anticipating*. Before round t , it ranks the actions by how often they have been selected in the previous rounds. The $\lceil K/2 \rceil$

most frequently selected actions receive reward 0, while the remaining actions receive reward 1. Ties are resolved by a deterministic rotating order.

Thus r_t can depend on the learner's past actions, but not on the fresh random draw I_t from the current round. For the main $K = 10$ experiments, five actions are punished each round. HFA is used both for the main one-player comparison and for the action-space study.

4.2.2 Oblivious Lazy Random Walk Environment

The lazy random walk (LRW) provides an *oblivious* contrast: its reward sequence is generated independently of the learner's actions. Each action has its own process on

$$\{0, 0.1, \dots, 0.9, 1\},$$

starting at 0.5. At an interior state, the next reward moves down by 0.1, stays unchanged, or moves up by 0.1, each with probability $1/3$. At either boundary, it stays or moves one step inward, each with probability $1/2$.

For a fixed replicate, the complete reward paths are generated independently of the learner. Algorithms compared within that replicate therefore face the same underlying LRW realization. In contrast to HFA, the learner's actions do not affect future rewards.

4.3 Repeated Games

The repeated-game experiments use Rock–Paper–Scissors (RPS) and Rock–Paper–Scissors–Lizard–Spock (RPSLS). Raw win/tie/loss payoffs $1, 0, -1$ are normalized to $1, 0.5, 0$, so both games are symmetric constant-sum games with total payoff 1 at every action profile.

For RPS, with actions ordered Rock, Paper, Scissors, player 0's payoff matrix is

$$U_{\text{RPS}} = \begin{pmatrix} 0.5 & 0 & 1 \\ 1 & 0.5 & 0 \\ 0 & 1 & 0.5 \end{pmatrix}.$$

Player 1 receives $1 - U_{\text{RPS}}$ entrywise.

For RPSLS, with actions ordered Rock, Paper, Scissors, Lizard, Spock, player 0's payoff matrix is

$$U_{\text{RPSLS}} = \begin{pmatrix} 0.5 & 0 & 1 & 1 & 0 \\ 1 & 0.5 & 0 & 0 & 1 \\ 0 & 1 & 0.5 & 1 & 0 \\ 0 & 1 & 0 & 0.5 & 1 \\ 1 & 0 & 1 & 0 & 0.5 \end{pmatrix},$$

and again player 1 receives the complementary payoff $1 - U_{\text{RPSLS}}$.

At each round, both players sample their actions before feedback is produced. Under full information, a player observes the payoff of every one of its actions against the opponent's realized action. Under bandit feedback, it observes only the payoff of its played action. The

counterfactual payoff vector is still retained by the simulator for offline regret evaluation and is never shown to a bandit learner.

RPS is mainly used to inspect the structure of the empirical joint-action distribution. RP-SLS is the main game for self-play, horizon-scaling, and cross-play comparisons.

4.4 Evaluation Metrics

4.4.1 Expected Action Regret

Following the terminology of Greenwald et al. [10], the evaluator computes *action regret* from the actions actually sampled in each replicate. It maintains the replacement gains

$$G_T(i, j) = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)).$$

External, internal, and swap action regret are then

$$R_T^{\text{ext}} = \max_j \sum_i G_T(i, j), \quad R_T^{\text{int}} = \max_{i,j} G_T(i, j),$$

and

$$R_T^{\text{swap}} = \sum_i \max_j G_T(i, j).$$

The reported trajectories are arithmetic means over the 12 replicates and are therefore Monte Carlo estimates of the corresponding expected action regret,

$$\mathcal{R}_T^q = \mathbb{E}[R_T^q], \quad q \in \{\text{ext}, \text{int}, \text{swap}\}.$$

For swap regret, this placement of the expectation follows the expected swap regret notion distinguished from pseudo-regret by Huang and Pan [12]. The main plots use R_t/t ; selected supplementary plots use $R_t/\sqrt{t}$. The same evaluator is used for all algorithms under both feedback models.

4.4.2 Equilibrium Distance

For repeated games, the empirical joint distribution after t rounds is

$$\hat{\mu}_t(a) = \frac{1}{t} \sum_{s=1}^t \mathbf{1}\{a_s = a\}.$$

The experiments compute its full-dimensional L_1 distance to the complete CE and CCE sets:

$$d_E(\hat{\mu}_t) = \min_{\mu \in E} \|\hat{\mu}_t - \mu\|_1, \quad E \in \{\text{CE}, \text{CCE}\}.$$

The corresponding linear programs are solved with HiGHS. Distances are computed separately for each replicate and then averaged. Joint-action heatmaps instead average the final empirical distributions across replicates.

4.5 Finite-Horizon Scaling

For each independently configured horizon, the final action regret is first averaged across replicates. Positive replicate-mean endpoints are fitted by ordinary least squares in log–log coordinates,

$$\log R_T = \alpha \log T + \log c, \quad \text{equivalently} \quad R_T \approx cT^\alpha.$$

The fitted α and c are used only as finite-horizon empirical descriptors, not as estimates of asymptotic regret rates.

Trajectory figures show replicate means without confidence bands. Endpoint information pages additionally report the sample standard deviation and standard error. These statistics are descriptive and are not used for formal hypothesis testing.

5 Experimental Results

This chapter reports the empirical behavior of the algorithms introduced in Chapter 3. The presentation is organized around recurring patterns rather than around the complete set of generated plots. The one-player experiments show how the three regret notions respond to changes in the reward process, the action-space size, and the horizon; the repeated-game experiments add self-play, cross-play, and empirical CE/CCE behavior. Supplementary trajectory views, individual horizon-scaling fits, selected action-space examples, and selected repeated-game profiles are collected in Appendix A.

Section 5.1 covers the historical-frequency and lazy-random-walk reward processes together with the action-space and horizon-scaling studies. Section 5.2 then considers self-play and cross-play in RPSLS and RPS. The discussion in this chapter remains descriptive; the connection to the theoretical results from earlier chapters is taken up in Chapter 6. Unless stated otherwise, the comparative statements below refer to the tested environments and game instances rather than to the corresponding classes in general. The plotted curves and endpoint comparisons are replicate means; small differences between nearby endpoints are therefore treated descriptively rather than as evidence of a statistically established ranking.

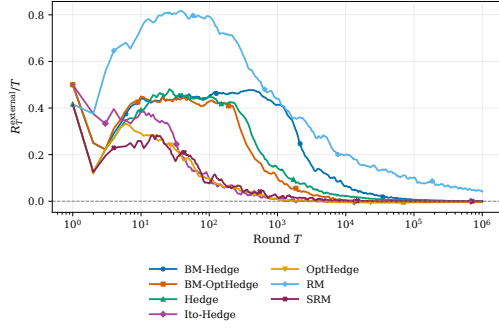
5.1 One-Player Regret Experiments

This section compares the one-player experiments across reward processes, action-space sizes, and horizons. The main text focuses on representative comparisons, while supplementary trajectory views, individual horizon-scaling fits, and selected action-space examples are collected in Appendix A.1. Unless stated otherwise, regret trajectories use R_T/T , i.e., average regret per round; supplementary $R_T/\sqrt{T}$ views in Appendix A.1.1 are used where this normalization makes finite-horizon differences easier to distinguish.

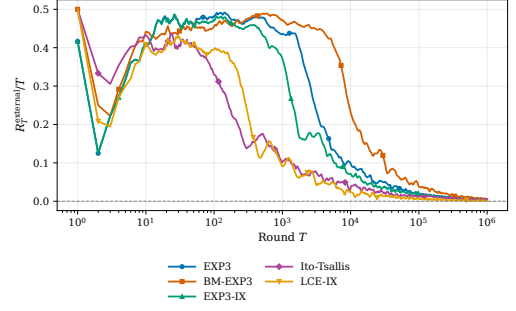
5.1.1 Comparison Across Environments

The two reward processes exhibit markedly different degrees of separation among the tested learners. The historical-frequency environment provides the higher-contrast case, whereas many of these differences are compressed in the lazy random walk. We therefore begin with the historical-frequency results before comparing the corresponding behavior under the lazy-random-walk environment.

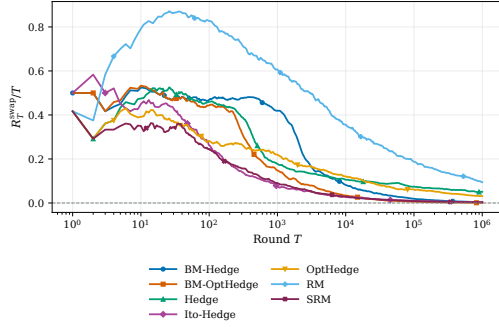
Historical-frequency environment. Figure 5.1 compares external and swap regret under both feedback models.



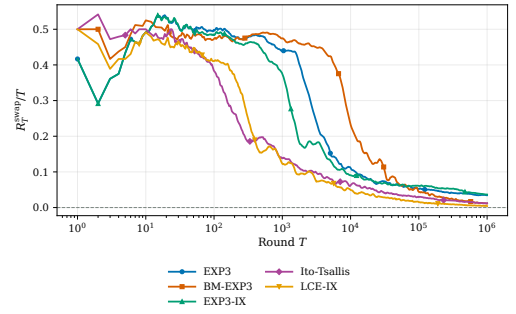
(a) Full information, external regret



(b) Bandit, external regret



(c) Full information, swap regret



(d) Bandit, swap regret

Figure 5.1: External and swap regret in the historical-frequency environment.

Under full information, most algorithms become tightly concentrated under the external benchmark. RM is the main exception, while the remaining learners occupy a comparatively narrow region toward the end of the run. Under swap regret, the broad separation is already visible in Figure 5.1, although the late-horizon differences are compressed by the $R_T/\sqrt{T}$ normalization. The supplementary $R_T/\sqrt{T}$ view in Figure A.1 makes the grouping clearer: Hedge and OptHedge remain well above BM-Hedge, Ito-Hedge, SRM, and BM-OptHedge.

The bandit comparison shows the same broad contrast. The external-regret trajectories remain comparatively close, whereas under swap regret EXP3 and EXP3-IX form the upper group, BM-EXP3 and Ito-Tsallis occupy an intermediate range, and LCE-IX lies below them.

Thus, the empirical grouping of the algorithms changes with the regret notion; the differences visible under swap regret are not reproduced by a simple rescaling of the external-regret plot.

Lazy-random-walk environment. Figure 5.2 shows the corresponding external- and swap-regret comparisons for the lazy-random-walk environment. The separation among the tested learners is considerably smaller than in the historical-frequency case.

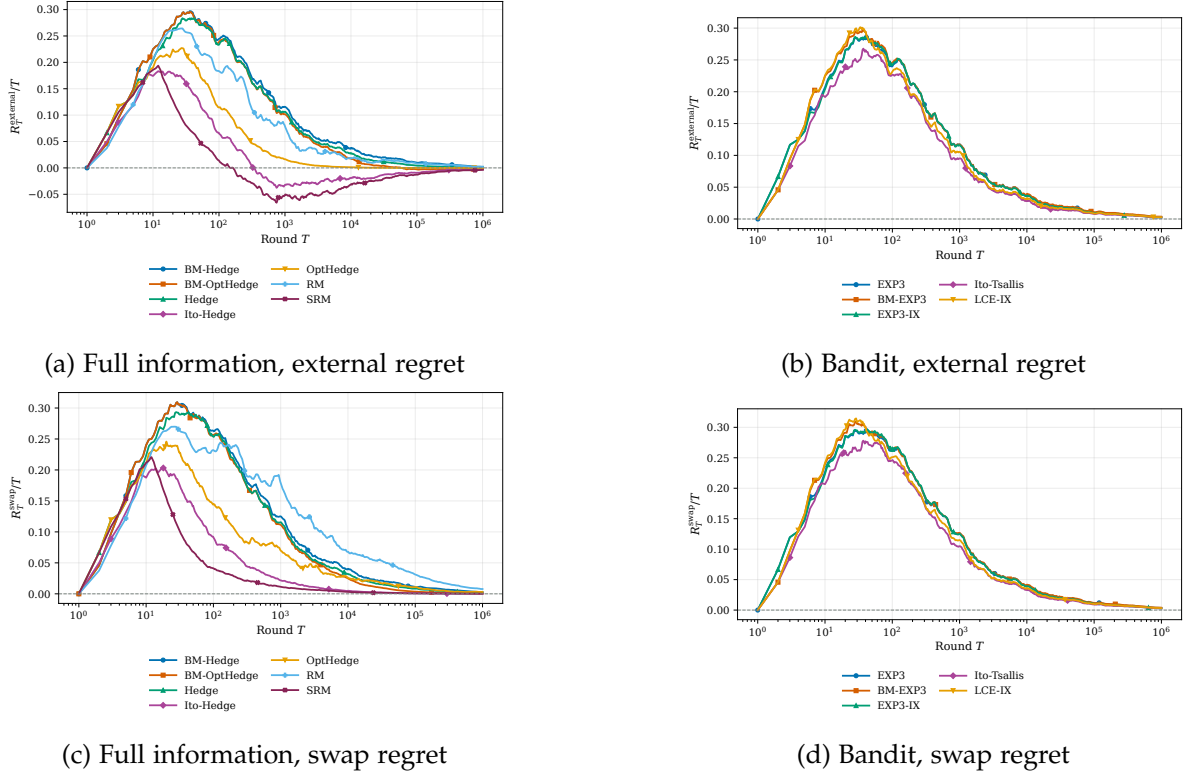


Figure 5.2: External and swap regret in the lazy-random-walk environment.

Under full information, the swap-regret curves are much more tightly grouped than in the historical-frequency environment. This late-horizon ordering is again clearer under the $R_T/\sqrt{T}$ normalization in Figure A.1: Hedge, OptHedge, and BM-Hedge occupy a similar range, while Ito-Hedge, SRM, and BM-OptHedge remain near the lower end of the comparison. RM remains visibly above the other methods.

The compression is even more striking under bandit feedback. EXP3, EXP3-IX, BM-EXP3, Ito-Tsallis, and LCE-IX remain close under external regret and also finish in a narrow range under swap regret. The three-level grouping visible in the historical-frequency experiment largely disappears.

The contrast with HFA is not a uniform shift in regret magnitude. Across the two tested reward processes, both the amount of separation and several pairwise orderings change.

5.1.2 Relations Among Regret Notions

The aggregate plots compare algorithms one regret notion at a time. Viewing all three notions for the same learner exposes a complementary feature of the data: internal regret often evolves differently from external and swap regret. Figure 5.3 illustrates this for Hedge and EXP3 in both environments.

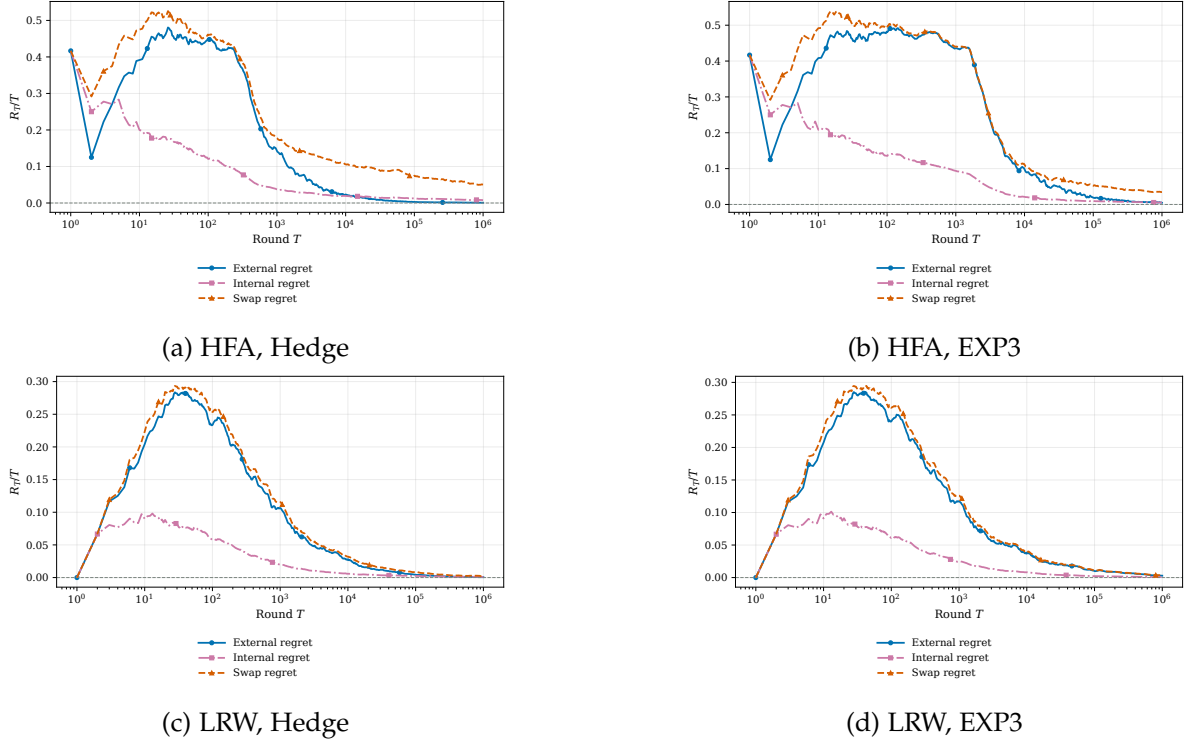


Figure 5.3: External, internal, and swap regret for Hedge and EXP3 in both one-player environments.

A recurring visual feature is that external and swap regret often share more of their large-scale temporal movement than either shares with internal regret. Internal regret can occupy a distinctly different range and may flatten, rise, or decline differently over the same interval.

The similarity concerns shape rather than scale. External and swap regret can move in parallel while remaining more than an order of magnitude apart, and comparable endpoint values do not imply similar trajectories over the run.

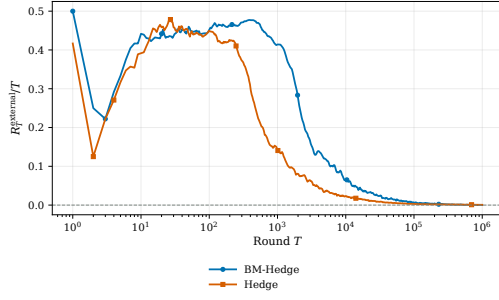
The same broad distinction also appears in the action-space experiments. Across multiple values of K , both Hedge and EXP3 show similar large-scale temporal phases in external and swap regret, whereas internal regret follows a visibly different trajectory (Figure 5.8).

5.1.3 Pairwise Algorithm Comparisons

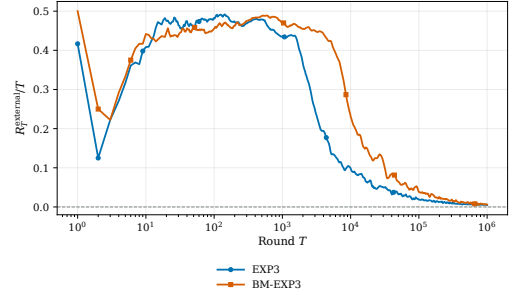
The aggregate figures identify broad groups of algorithms, but pairwise comparisons make differences in trajectory timing and late-horizon ordering easier to distinguish. Unless stated otherwise, the comparisons below use the historical-frequency environment.

Direct and BM-reduced learners. Figure 5.4 compares the standalone external-regret learners with their BM reductions across all three regret notions.

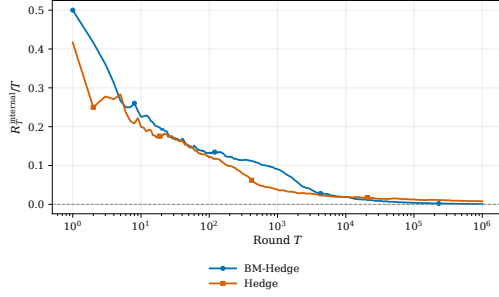
5 Experimental Results



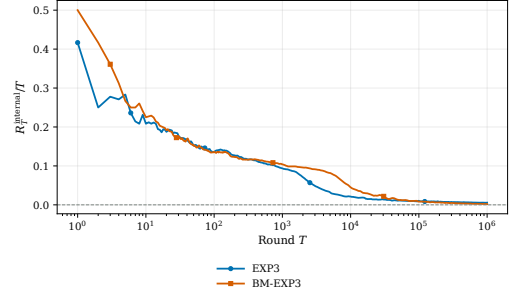
(a) Hedge and BM-Hedge, external regret



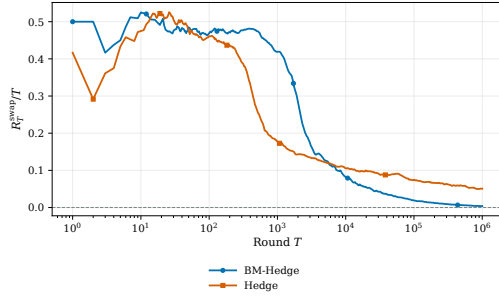
(b) EXP3 and BM-EXP3, external regret



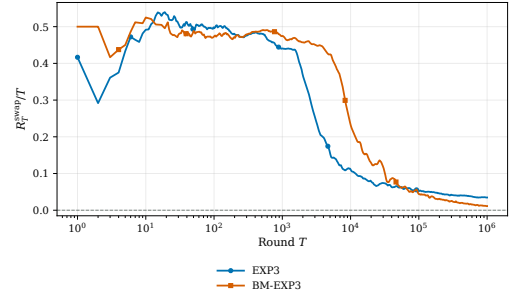
(c) Hedge and BM-Hedge, internal regret



(d) EXP3 and BM-EXP3, internal regret



(e) Hedge and BM-Hedge, swap regret

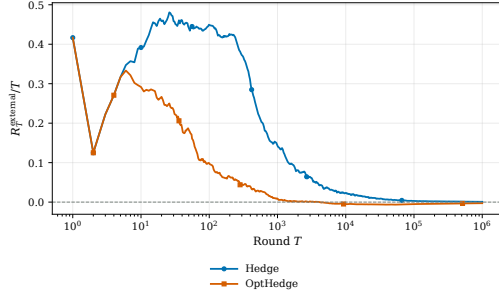


(f) EXP3 and BM-EXP3, swap regret

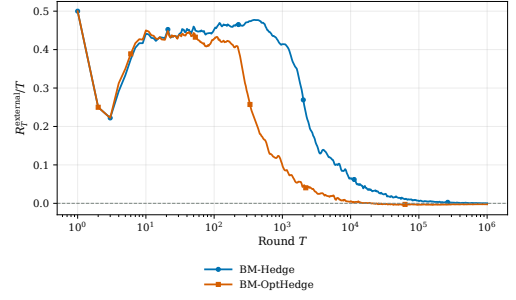
Figure 5.4: Regret comparison of direct learners and their BM reductions in the historical-frequency environment.

Under external regret, both direct/BM pairs remain relatively close. Under internal and swap regret, the standalone learner begins to decline earlier, but the BM reduction later crosses below it. This pattern appears for both Hedge / BM-Hedge and EXP3 / BM-EXP3, with the largest late-horizon separation under swap regret.

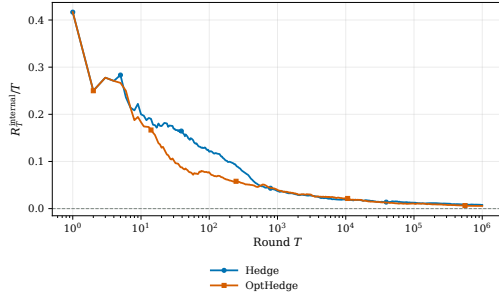
Optimistic variants. Figure 5.5 compares the effect of optimism for the standalone Hedge learner and inside the BM reduction across all three regret notions.



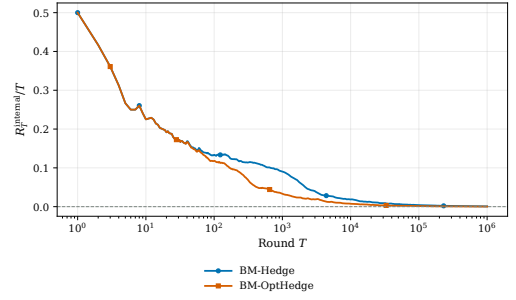
(a) Hedge and OptHedge, external regret



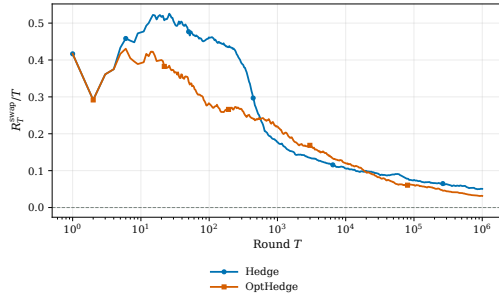
(b) BM-Hedge and BM-OptHedge, external regret



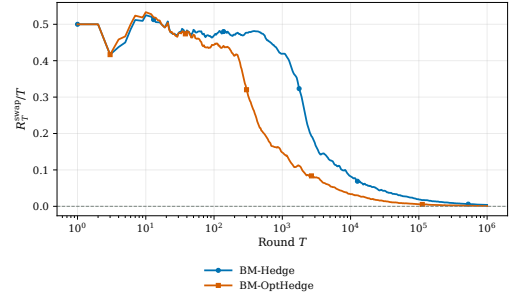
(c) Hedge and OptHedge, internal regret



(d) BM-Hedge and BM-OptHedge, internal regret



(e) Hedge and OptHedge, swap regret



(f) BM-Hedge and BM-OptHedge, swap regret

Figure 5.5: Hedge and OptHedge, with and without BM reduction, in the historical-frequency environment.

For both pairs, the optimistic variant begins to decline earlier and reaches a lower late-horizon regret level than its non-optimistic counterpart. The separation is modest for Hedge / OptHedge but substantially larger within the BM reduction, particularly under internal and swap regret.

EXP3 versus EXP3-IX. Figure 5.6 compares EXP3 and EXP3-IX under both the R_T/T and $R_T/\sqrt{T}$ normalizations in the historical-frequency environment.

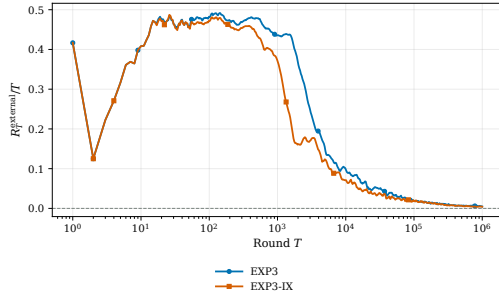
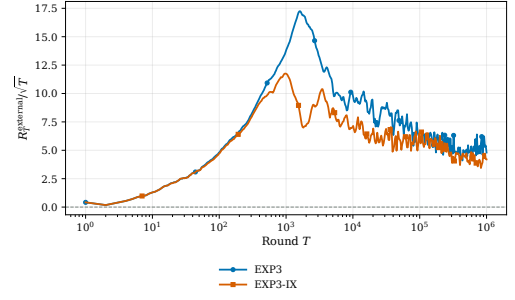
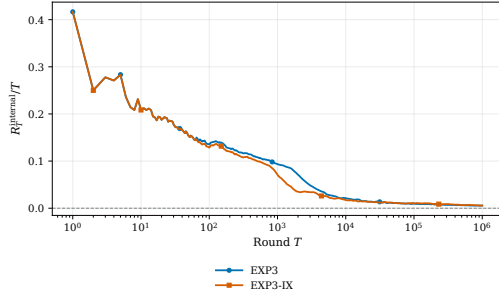
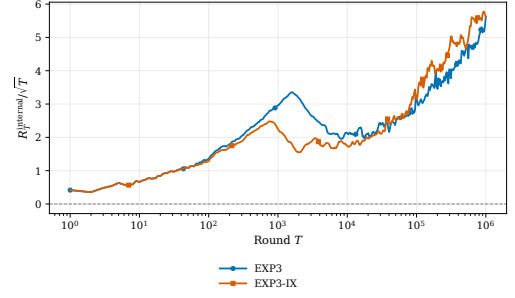
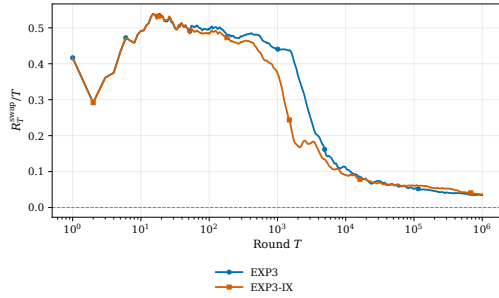
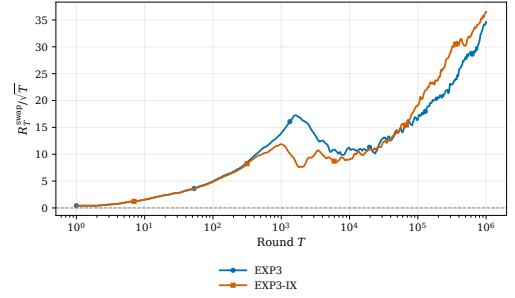
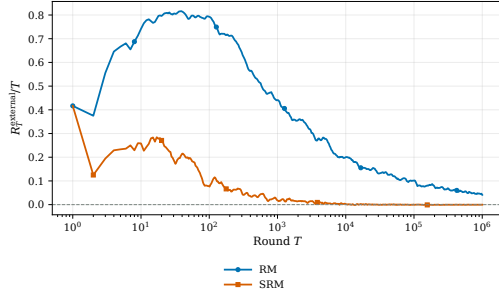

 (a) External regret, R_T/T

 (b) External regret, $R_T/\sqrt{T}$

 (c) Internal regret, R_T/T

 (d) Internal regret, $R_T/\sqrt{T}$

 (e) Swap regret, R_T/T

 (f) Swap regret, $R_T/\sqrt{T}$

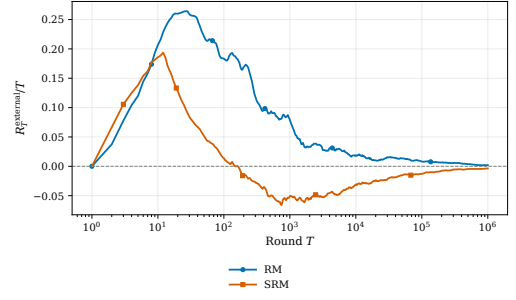
 Figure 5.6: EXP3 and EXP3-IX in the historical-frequency environment under the R_T/T and $R_T/\sqrt{T}$ normalizations.

EXP3-IX begins to decline earlier than EXP3 across all three regret notions. The late-horizon ordering is not uniform across the metrics: EXP3-IX remains lower under external regret, the internal-regret curves finish at nearly the same level, and the swap-regret curves cross, with EXP3 ending lower. The $R_T/\sqrt{T}$ panels preserve the same ordering as the R_T/T panels but make the late-horizon differences more visible.

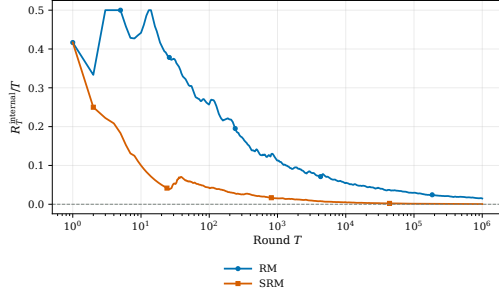
RM versus SRM. Figure 5.7 compares RM and SRM across all three regret notions in both one-player environments.



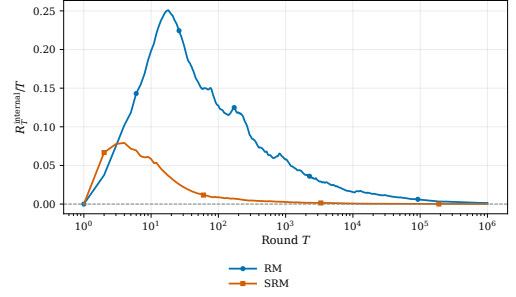
(a) HFA, external regret



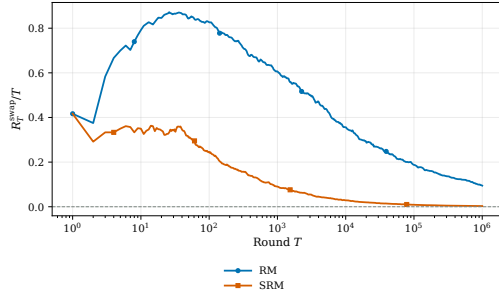
(b) LRW, external regret



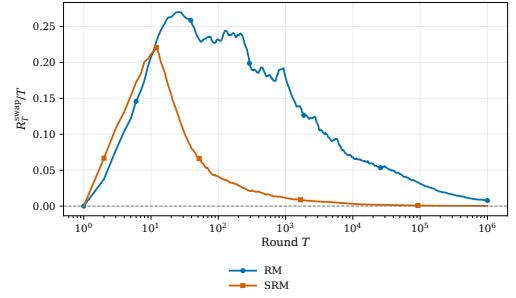
(c) HFA, internal regret



(d) LRW, internal regret



(e) HFA, swap regret



(f) LRW, swap regret

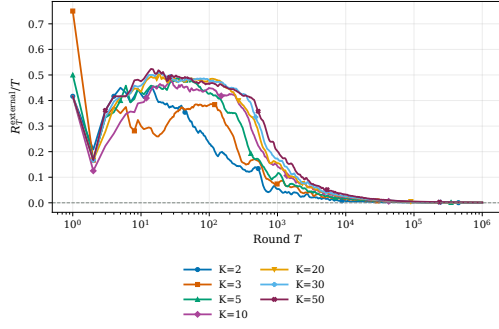
Figure 5.7: RM and SRM in both one-player environments.

SRM remains substantially below RM under all three regret notions in both one-player environments, and the separation remains large over the later part of the horizon. Supplementary $R_T/\sqrt{T}$ views are shown in Figure A.2; the corresponding independent-horizon growth fits are discussed in Section 5.1.5.

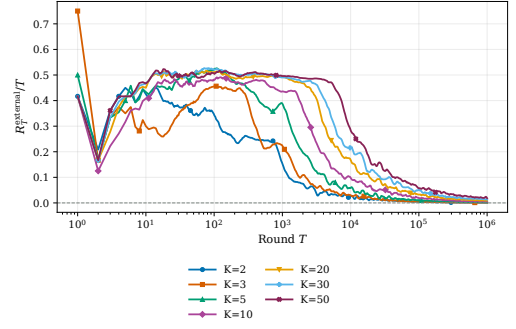
5.1.4 Action-Space Scaling

The historical-frequency experiment was repeated for $K \in \{2, 3, 5, 10, 20, 30, 50\}$. Figure 5.8 shows the resulting trajectories for Hedge and EXP3 across all three regret notions.

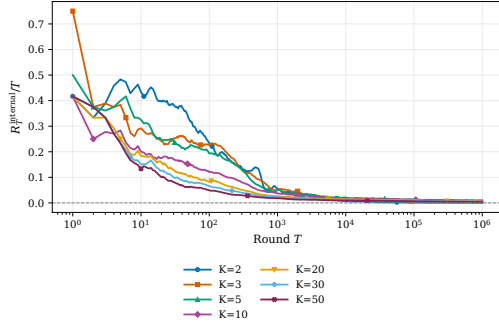
5 Experimental Results



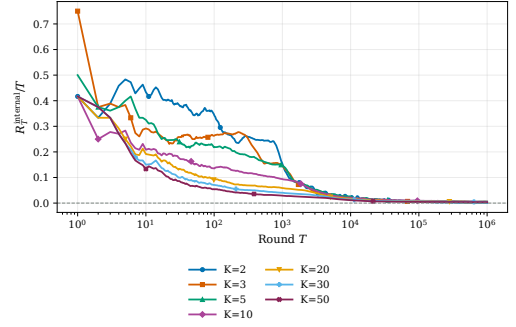
(a) Hedge, external regret



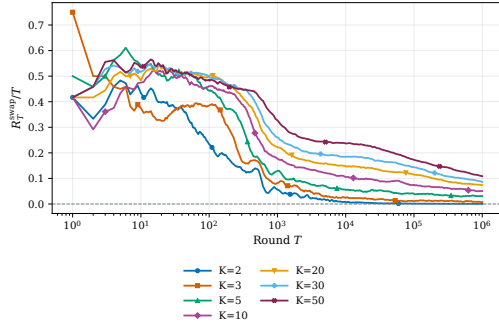
(b) EXP3, external regret



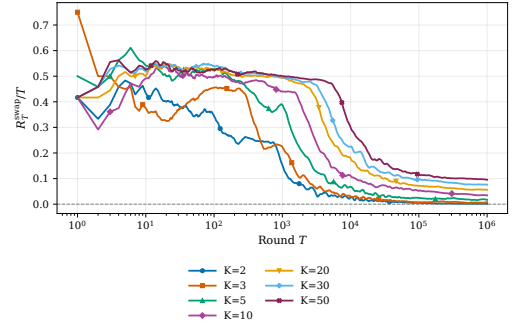
(c) Hedge, internal regret



(d) EXP3, internal regret



(e) Hedge, swap regret



(f) EXP3, swap regret

Figure 5.8: Hedge and EXP3 across action-space sizes in the historical-frequency environment.

A common pattern under external and swap regret is an initial plateau followed by a pronounced decline. This is particularly clear for EXP3 and for the supplementary BM-Hedge, BM-EXP3, and Ito-Tsallis examples in Appendix A.1.3, and less pronounced for Hedge. Increasing K extends this plateau, so that the main decline begins later.

Internal regret shows the opposite dependence on K : for Hedge and EXP3, larger action spaces tend to enter the declining phase earlier. Thus, increasing K delays the decline under external and swap regret, but shifts it earlier under internal regret.

For both displayed learners, the endpoint swap-regret levels also increase strongly with K .

For Hedge, the final value rises from approximately 4.3×10^{-4} at $K = 2$ to 0.109 at $K = 50$, and for EXP3 from about 0.0012 to 0.096.

5.1.5 Horizon Scaling

The experiments were repeated at several independently configured horizons. For each horizon, the learner was reinitialized and the final cumulative regret was fitted by

$$R_T \approx cT^\alpha.$$

The exponent α is used only as a descriptive finite-horizon statistic, not as an estimate of the asymptotic convergence rate. The individual log–log fits are shown in Appendix A.1.2.

Table 5.1: Empirical horizon-scaling exponents for external, internal, and swap regret in the two one-player environments.

Algorithm	Historical-frequency			Lazy random walk		
	Ext.	Int.	Swap	Ext.	Int.	Swap
<i>Full information</i>						
Hedge	0.478	0.757	0.766	0.495	0.493	0.491
OptHedge	–	0.729	0.743	–	0.525	0.526
BM-Hedge	–	0.493	0.524	0.518	0.507	0.506
BM-OptHedge	–	0.378	0.357	–	0.321	0.291
Ito-Hedge	–	0.534	0.546	–	0.364	0.301
RM	0.659	0.708	0.726	0.558	0.471	0.567
SRM	–	0.490	0.494	–	0.382	0.391
<i>Bandit feedback</i>						
EXP3	0.481	0.705	0.716	0.536	0.525	0.528
EXP3-IX	0.546	0.738	0.764	0.518	0.514	0.513
BM-EXP3	0.433	0.514	0.522	0.533	0.517	0.527
Ito-Tsallis	0.534	0.622	0.653	0.502	0.509	0.502
LCE-IX	0.375	0.525	0.515	0.516	0.513	0.514

Note: “–” indicates that the log–log fit is unavailable because the replicate-mean regret is non-positive at one or more required fit points.

The historical-frequency environment shows the clearest separation between regret notions. For Hedge, EXP3, and EXP3-IX, the external-regret exponent is close to 0.5, while the internal- and swap-regret exponents are around 0.7–0.76. Several stronger-regret methods exhibit substantially smaller internal- and swap-regret exponents; for example, BM-Hedge has 0.493 and 0.524, while BM-OptHedge has 0.378 and 0.357.

In the lazy-random-walk environment, these differences are much less pronounced. For most learners, the three fitted exponents cluster near 0.5. The horizon-scaling experiment therefore reinforces the main environment contrast observed in the trajectories: growth rates are more heterogeneous in HFA and more compressed in LRW.

5.2 Repeated-Game Self-Play and Cross-Play

The repeated-game experiments add two features that are absent from the one-player reward sequences: the empirical joint distribution can be compared with the CE and CCE sets, and a learner can be evaluated against different opponents. RPSLS is used as the main self-play and cross-play benchmark, while RPS is used where its smaller joint-action space makes temporal and joint-distribution structure easier to see. Table 5.2 summarizes the finite-horizon regret growth observed in RPSLS self-play.

Table 5.2: Empirical horizon-scaling parameters for RPSLS self-play. For each regret notion, the replicate-mean final cumulative action regret is fitted by $R_T \approx cT^\alpha$ over independently configured horizons from 10^3 to 10^6 . The table reports player 0; a fixed player is used because the parameters are obtained from separate regressions rather than averaged across players.

<i>Full information</i>							
	Hedge	OptHedge	BM-Hedge	BM-OptHedge	Ito-Hedge	RM	SRM
α_{ext}	0.474	0.524	0.480	0.522	0.454	0.753	0.546
c_{ext}	0.8150	0.07257	0.6491	0.3338	0.9253	0.5571	0.5753
α_{int}	0.915	0.578	0.518	0.492	0.508	0.692	0.522
c_{int}	0.05204	0.5963	0.4369	0.5619	0.5081	0.6543	0.4048
α_{swap}	0.964	0.573	0.499	0.506	0.506	0.708	0.533
c_{swap}	0.1345	2.172	1.247	1.135	1.182	1.704	1.138

<i>Bandit feedback</i>					
	EXP3	EXP3-IX	BM-EXP3	Ito-Tsallis	LCE-IX
α_{ext}	0.530	0.561	0.629	0.570	0.581
c_{ext}	1.691	1.113	0.5249	1.075	0.7204
α_{int}	0.627	0.642	0.584	0.570	0.541
c_{int}	0.3451	0.2697	0.3750	0.4885	0.5289
α_{swap}	0.680	0.684	0.606	0.579	0.556
c_{swap}	0.5670	0.5186	0.7782	1.200	1.152

5.2.1 Regret Scaling in Self-Play

The full-information fits show that the main separation appears only after the benchmark is strengthened. Except for RM, the external-regret exponents are clustered around the square-root scale. Under internal and swap regret, however, BM-Hedge, BM-OptHedge, Ito-Hedge, and SRM remain close to $\alpha = 0.5$, whereas Hedge rises to 0.915 and 0.964. OptHedge lies between these groups, and RM remains distinctly higher. The separation between the algorithms is more pronounced for internal and swap regret than for external regret.

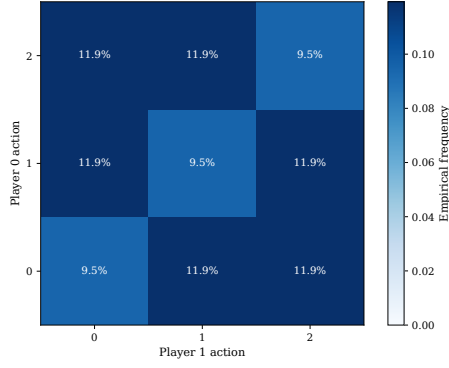
The bandit setting shows the same pattern more weakly. The external-regret exponents occupy a relatively narrow range, while EXP3 and EXP3-IX move above 0.62 under internal and swap regret. BM-EXP3, Ito-Tsallis, and LCE-IX remain lower, with internal- and swap-regret exponents between roughly 0.54 and 0.61. The separation is therefore more compressed than under full information, but it again becomes clearer under the stronger regret notions.

The fitted constant c is important for the finite-horizon scale and prevents α from being read as an endpoint ranking. For example, Hedge combines large internal- and swap-regret exponents with relatively small fitted constants, while OptHedge has smaller exponents but a much larger swap-regret constant. The table is therefore used to compare growth profiles rather than to rank algorithms by a single fitted parameter.

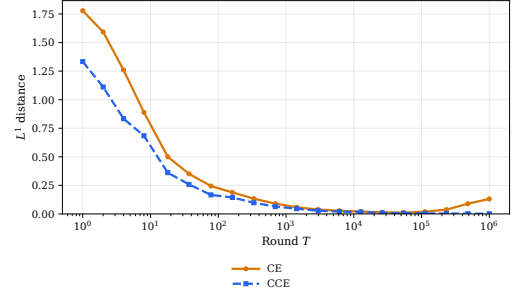
The corresponding R_T/T and $R_T/\sqrt{T}$ trajectories at $T = 10^6$ are collected in Appendix A.2.1. These panels visualize the within-run behavior at the largest horizon and are supplementary to the independent-horizon regressions summarized above.

5.2.2 Joint-Action Structure and Equilibrium Distance

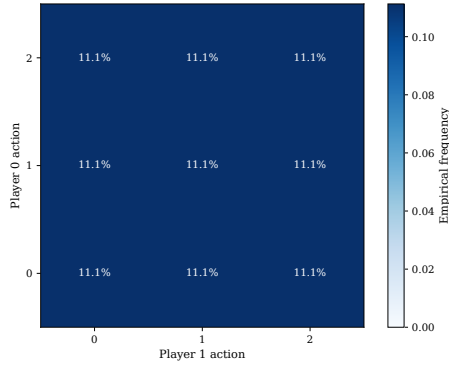
The regret fits above do not by themselves describe the empirical joint distribution. In the equilibrium results, low CCE distance can still coexist with much larger CE distance, most clearly for Hedge and, more weakly, for OptHedge. RPS provides a cleaner view of the structure behind this difference because its joint-action distribution contains only nine cells.



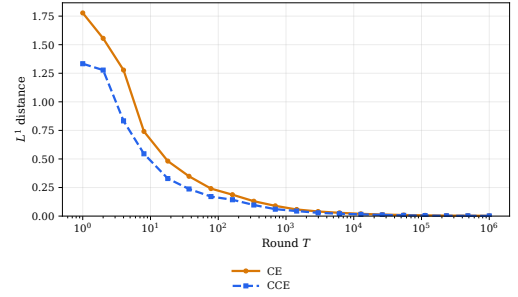
(a) Hedge, joint actions



(b) Hedge, CE/CCE distance



(c) BM-Hedge, joint actions



(d) BM-Hedge, CE/CCE distance

Figure 5.9: Joint-action distributions and CE/CCE distances for Hedge and BM-Hedge self-play in RPS.

Hedge has nearly uniform marginal play, yet its joint distribution retains a regular diagonal/off-diagonal distortion: the diagonal cells contain about 9.5% of the mass each, compared with about 11.9% for the off-diagonal cells. BM-Hedge is nearly uniform at the displayed precision. The difference is therefore not visible from marginal frequencies alone; it appears in how the two players' actions are correlated over time.

The equilibrium trajectories also reveal differences that are not visible from the endpoint values alone. For Hedge, CE and CCE distance initially decrease together and become nearly indistinguishable over an intermediate interval. The CCE distance then remains near zero, while the CE distance turns upward and separates late. The final CE–CCE gap is therefore created by a change in the trajectory rather than by a persistent gap throughout the run. A weaker late separation is also observed for EXP3 and EXP3-IX, and the same qualitative effect appears for Hedge in RPSLS. Selected supplementary joint-action distributions and equilibrium trajectories are shown in Appendix A.2.2.

5.2.3 Opponent Dependence in Asymmetric Cross-Play

The same horizon-scaling analysis can be used to compare cross-play profiles. Table 5.3 reports the fitted parameters separately for both players. Profiles are written as player 0 versus player 1.

Table 5.3: Empirical horizon-scaling parameters for selected RPSLS cross-play profiles. For each player and regret notion, the replicate-mean final cumulative action regret is fitted by $R_T \approx cT^\alpha$ over independently configured horizons from 10^3 to 10^6 .

Full information						
	Hedge–Hedge	Hedge–BM–Hedge	Hedge–Ito–Hedge	Hedge–OptHedge	Hedge–SRM	RM–SRM
Player 0						
α_{ext}	0.474	0.479	0.431	0.510	0.513	0.700
c_{ext}	0.8150	0.6272	1.042	0.5083	0.5279	1.766
α_{int}	0.915	0.516	0.527	0.664	0.574	0.682
c_{int}	0.05204	0.4949	0.5137	0.3537	0.3538	0.9542
α_{swap}	0.964	0.510	0.533	0.647	0.574	0.698
c_{swap}	0.1345	1.260	1.163	1.425	0.8454	2.665
Player 1						
α_{ext}	0.499	0.520	0.552	0.563	0.490	–
c_{ext}	0.5800	0.4600	0.3855	0.2218	0.8104	–
α_{int}	0.916	0.505	0.542	0.698	0.500	0.343
c_{int}	0.05179	0.5496	0.4087	0.2898	0.4584	0.5434
α_{swap}	0.971	0.519	0.550	0.667	0.505	0.359
c_{swap}	0.1236	1.060	0.9190	1.246	1.363	1.160
Bandit feedback						
	EXP3–EXP3	EXP3–BM–EXP3	EXP3–EXP3–IX	EXP3–Ito–Tsallis		
Player 0						
α_{ext}	0.530	0.585	0.531	0.535		
c_{ext}	1.691	0.8716	1.689	1.457		
α_{int}	0.627	0.563	0.616	0.551		
c_{int}	0.3451	0.4879	0.3720	0.6033		
α_{swap}	0.680	0.575	0.642	0.566		
c_{swap}	0.5670	1.168	0.8066	1.359		
Player 1						
α_{ext}	0.544	0.543	0.539	0.566		
c_{ext}	1.430	1.673	1.296	1.152		
α_{int}	0.638	0.542	0.616	0.575		
c_{int}	0.3091	0.7275	0.3404	0.4950		
α_{swap}	0.687	0.552	0.657	0.583		
c_{swap}	0.5247	1.711	0.6419	1.172		

Note: “–” indicates that the log-log fit is unavailable because the replicate-mean regret is non-positive at one or more required fit points.

The strongest opponent effect appears for Hedge under the stronger regret notions. In self-play, the internal- and swap-regret exponents are close to one for both players. Against BM-Hedge, Ito-Hedge, or SRM, they instead move close to the 0.5 range; pairing with OptHedge gives an intermediate pattern. Thus, changing the opponent alters the finite-horizon growth profile rather than merely shifting the regret curves by a constant factor.

The bandit profiles show the same effect more weakly. EXP3 has larger internal- and swap-regret exponents in self-play and against EXP3-IX than against BM-EXP3 or Ito-Tsallis, while the corresponding player-1 fits follow a similar but not identical ordering. Heterogeneous pairing therefore does not produce a uniform improvement.

The RM-SRM pairing deserves a separate comparison because the effect is strongly asymmetric. Table 5.4 applies the same horizon-scaling analysis to RM-RM, RM-SRM, and SRM-SRM. In the mixed profile, player 0 uses RM and player 1 uses SRM.

Table 5.4: Empirical horizon-scaling parameters for the RM/SRM RPSLS profiles. For each player and regret notion, the replicate-mean final cumulative action regret is fitted by $R_T \approx cT^\alpha$ over independently configured horizons from 10^3 to 10^6 .

	RM-RM	RM-SRM	SRM-SRM
<i>Player 0</i>			
α_{ext}	0.753	0.700	0.546
c_{ext}	0.5571	1.766	0.5753
α_{int}	0.692	0.682	0.522
c_{int}	0.6543	0.9542	0.4048
α_{swap}	0.708	0.698	0.533
c_{swap}	1.704	2.665	1.138
<i>Player 1</i>			
α_{ext}	0.691	—	0.501
c_{ext}	1.133	—	0.6392
α_{int}	0.655	0.343	0.498
c_{int}	1.007	0.5434	0.4224
α_{swap}	0.675	0.359	0.503
c_{swap}	2.453	1.160	1.238

Note: “—” indicates that the log-log fit is unavailable because the replicate-mean regret is non-positive at one or more required fit points.

The mixed profile is especially favorable to SRM. When player 1 uses SRM against RM, its internal- and swap-regret exponents fall to 0.343 and 0.359, compared with 0.498 and 0.503 in SRM–SRM self-play. The fitted coefficients remain of comparable order. On the RM side, the corresponding exponents change only slightly from RM–RM to RM–SRM, while the fitted coefficients increase. The mixed profile therefore does not interpolate symmetrically between the two self-play cases. In the tested RPSLS interaction, SRM appears to benefit from facing RM and exhibits a distinctly more favorable stronger-regret growth profile.

5.3 Summary of Findings

Overall, the experiments show that finite-horizon behavior depends strongly on the regret notion, environment, and opponent. The horizon- and action-space experiments further show that these differences persist across problem scales and are not captured by endpoint values alone. In repeated games, regret and equilibrium trajectories provide complementary views of the learning dynamics. Chapter 6 discusses these findings in relation to the theoretical results and the limitations of the experimental setting.

6 Discussion

The experiments in Chapter 5 show that the comparison between the learners depends strongly on the regret notion and the experimental setting. This chapter returns to the research questions from Chapter 1: how the algorithms differ across the three regret notions, how stable these differences are across experimental settings, and how they appear in repeated games.

6.1 Regret Metrics and Theoretical Comparison

The experiments measure expected action regret, while the guarantees in Chapter 3 use several related regret notions. A few simple relations are enough to interpret most of the results.

First, action regret A_T and distribution regret D_T differ through the randomness of the sampled actions. In the bounded finite-action setting, this sampling difference is of root- T order,

$$A_T - D_T = O_p(\sqrt{T}),$$

up to logarithmic factors for the finite deviation class [10]. Thus a persistent super-root- T action-regret rate cannot be explained by sampling alone and implies the same leading polynomial order for distribution regret. Around the root- T scale, however, the two cannot be separated reliably by the current evaluator.

Second, several bandit guarantees concern pseudo-regret

$$P_T = \max_F \mathbb{E}[Z_F],$$

whereas the experiments estimate expected action regret

$$E_T = \mathbb{E} \left[\max_F Z_F \right].$$

Since

$$P_T \leq E_T,$$

sublinear expected action regret also implies sublinear pseudo-regret, but large expected action regret does not imply large pseudo-regret. This distinction is made explicit by Huang and Pan [12].

Some references give stronger probabilistic guarantees than the replicate means measured here. Huang and Pan derive an expected swap-regret bound from a high-probability instantaneous-regret bound [12]. Such high-probability statements are not tested directly by the mean regret curves used in the experiments.

These relations give a more precise interpretation of the individual algorithms.

Hedge. Its external action-regret exponents stay close to 0.5. Since action and distribution regret can differ by root- T sampling fluctuations, this does not determine the corresponding distribution-regret rate and therefore does not directly test the standard Hedge guarantee [3]. More interestingly, in RPSLS self-play its internal and swap exponents rise to about 0.92 and 0.96–0.97. If this trend persists, sampling noise cannot explain the difference: the corresponding distribution regrets must also grow faster than root- T .

OptHedge. The realized internal- and swap-regret exponents are above the root scale in several experiments. These quantities do not directly test the faster standalone OptHedge result of Chen and Peng [5], which concerns external mixed regret in optimistic self-play. The theoretical sub-root rate therefore remains unresolved by the current evaluator.

BM-Hedge. Its swap action-regret exponents are generally close to 0.5. Since this is also the scale of the action–distribution sampling difference, the experiment does not determine whether the corresponding distribution regret follows the root- T Blum–Mansour bound or a smaller rate [7].

BM-OptHedge. This method has some of the smallest internal- and swap-regret exponents in the one-player experiments. The empirical result is favorable, but the sub-root mixed-regret guarantee of Chen and Peng [5] cannot be identified directly from sampled action regret.

Ito-Hedge. Ito’s reduction gives a swap pseudo-regret guarantee [8]. The observed expected swap regret is sublinear in the tested settings, so $P_T \leq E_T$ also gives sublinear swap pseudo-regret if the observed trend persists. The exact pseudo-regret rate cannot be recovered from the current metric.

RM. RM shows sublinear internal and swap regret in the experiments, including its RPSLS profiles. This is compatible with the convergence behavior proved for the Hart–Mas-Colell repeated-game procedure [6]. That theorem is a self-play result, however, so the one-player and arbitrary cross-play experiments should mainly be viewed as empirical comparisons.

SRM. SRM also shows sublinear expected internal and swap regret, often with exponents near or below the root scale. This is consistent with its almost-sure pairwise-regret result under the assumptions of Hart and Mas-Colell [6], although expectation-level experiments cannot verify almost-sure convergence.

EXP3. Its expected external action regret remains roughly at the root scale, while internal and swap regret can grow substantially faster. Since the standard guarantee concerns external pseudo-regret [4], the sublinear external results also support sublinear pseudo-regret. The larger stronger-regret values do not contradict the external-regret guarantee.

EXP3-IX. Neu proves a high-probability bound for realized external regret [14]. The experiments instead report the mean realized regret across replicates. Its fitted external-regret exponents remain close to the root- T scale, but this expectation-level result does not establish the stronger high-probability guarantee.

BM-EXP3. Its expected swap regret is sublinear and usually close to the root scale. Because the literature guarantee is on swap pseudo-regret [7], this also supports sublinear pseudo-regret, although the fitted action-regret exponent should not be interpreted as the pseudo-regret exponent.

Ito-Tsallis. The same argument applies here. Expected swap regret remains sublinear in all tested environments, so the weaker swap pseudo-regret is also sublinear if the trend persists. The relatively larger exponent in HFA cannot be used to reject a sharper pseudo-regret bound.

LCE-IX. This is the cleanest comparison because Huang and Pan [12] directly bound expected swap action regret. The fitted swap-regret exponents are close to 0.5 in both one-player environments, matching the root- T dependence of the theoretical bound for fixed K . The experiment still does not test the stronger high-probability statement.

Overall, the empirical metric is informative in several directions. Large super-root action regret can rule out sampling as the explanation for a distributional effect, while small expected action regret can imply small pseudo-regret. Other comparisons remain unresolved, especially sub-root mixed-strategy rates and stronger probabilistic guarantees. These are limitations of the current evaluator rather than contradictions with the corresponding theory.

6.2 Robustness Across Experimental Conditions

The main empirical pattern is that the separation between algorithms is not equally strong in every environment. HFA produces much clearer differences in internal and swap regret than LRW, where many of the learners remain close together.

A plausible reason is the feedback structure of the two reward processes. LRW evolves independently of the learner, whereas HFA reacts to past action frequencies. An update that changes the learner’s behavior can therefore also change its future rewards in HFA. This feedback loop gives small differences between algorithms more opportunity to accumulate, which fits the stronger separation observed in the experiments.

The action-space experiments show a related finite-horizon effect. Increasing K generally raises swap regret and delays the main decline of the regret curves. The horizon fits likewise vary across environments: the exponent α alone does not determine which algorithm looks better at a practical horizon, since the coefficient c can shift the curves substantially.

The results therefore suggest a fairly simple picture. Algorithms designed for stronger regret usually become more clearly separated from external-regret baselines when internal

or swap regret is measured, but the size of this difference depends on the environment, action space, and horizon.

A broader difference is also visible between the two feedback settings. Under full information, several swap-regret-oriented methods show clearer separation from the external-regret baseline, and some achieve noticeably smaller fitted internal- and swap-regret exponents. Under bandit feedback, the corresponding differences are generally less pronounced, with most fitted exponents remaining closer to the root- T scale. This is consistent with the additional uncertainty introduced by observing only the reward of the sampled action. Since the two settings contain different algorithm families, however, this should be read as a qualitative comparison rather than a controlled feedback ablation.

6.3 Equilibrium and Opponent Effects in Repeated Games

Two patterns stand out in the repeated-game experiments.

First, the difference between regret notions is also visible in empirical equilibrium behavior. In RPS, Hedge self-play gets much closer to the CCE set than to the CE set, while several stronger-regret methods approach CE more closely. This matches the standard link between external regret and CCE, and between action-dependent regret and CE [6, 7].

The joint-action distributions make this intuitive. Hedge can have nearly uniform marginal action frequencies while still showing a structured joint distribution, so balanced marginals alone do not remove profitable conditional deviations.

The trajectories show a similar distinction. For Hedge, CE and CCE distance first decrease together; later, CCE distance stays near zero while CE distance rises again. A natural interpretation is that fixed deviations are already unprofitable while some action-dependent deviations remain. A weaker version of the same pattern appears for EXP3 and EXP3-IX.

Second, regret depends strongly on the opponent. Hedge has very large internal and swap exponents in self-play, but both move much closer to the root scale against BM-Hedge, Ito-Hedge, and SRM. EXP3 shows the same tendency more weakly under bandit feedback.

The RM-SRM profiles give the clearest asymmetric example. SRM has internal and swap exponents around 0.50 in SRM-SRM self-play, but they fall to about 0.34 and 0.36 against RM, while RM does not improve to the same extent. This suggests that the opponent can affect both regret levels and their finite-horizon growth.

Taken together, the repeated-game results support the expected connection between stronger regret and CE behavior, while showing that finite-horizon dynamics depend strongly on the interaction between the two learners.

7 Conclusion and Future Work

7.1 Conclusion

This thesis implemented and compared a range of regret-minimizing algorithms within a common experimental framework. The evaluation covered external, internal, and swap regret under both full-information and bandit feedback, as well as repeated-game behavior in RPS and RPSLS.

The experiments show that the choice of regret notion is important for an empirical comparison. Differences that are small under external regret can become much clearer once action-dependent deviations are considered. At the same time, the results depend on the reward process, action-space size, horizon, and, in repeated games, the opponent. The experiments therefore do not suggest a single ordering of the algorithms that applies across all tested settings.

The repeated-game experiments also connect these regret differences to equilibrium behavior. External regret and stronger regret notions can lead to different CE and CCE behavior, while cross-play shows that the observed regret of a learner can depend strongly on the learning dynamics of its opponent. Together with the finite-horizon scaling experiments, this provides an empirical view of distinctions that are less visible from asymptotic regret bounds alone.

The theoretical comparison in Chapter 6 also shows where the common action-regret evaluator is informative and where it is not. Relations between action regret, distribution regret, and pseudo-regret allow some empirical conclusions to be transferred between metrics, while sub-root mixed-strategy rates and stronger probabilistic guarantees cannot be resolved directly with the current evaluation.

7.2 Future Work

A natural next step is to add regret evaluators that more closely match the quantities used in the theoretical analyses. In particular, distribution and fully mixed regret would allow a more direct comparison with the guarantees for Hedge-based and optimistic methods. For high-probability results, replicate quantiles or empirical tail frequencies would complement the mean regret used here.

The repeated-game experiments could also be extended beyond the selected RPS and RPSLS profiles. A larger cross-play matrix and a broader set of games would make it possible to examine whether the opponent effects observed here persist in other settings. Equilibrium evaluation could also report the largest remaining profitable deviation, in addition to the

geometric distance to the CE and CCE sets.

Finally, longer horizons, more replicates, and larger action spaces would help separate persistent scaling behavior from finite-horizon effects. These extensions would use the same experimental framework while providing a more complete comparison between theoretical regret guarantees and observed learning behavior.

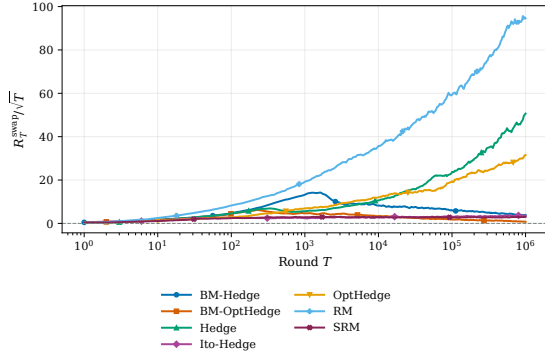
A Supplementary Experimental Material

This appendix collects supplementary experimental material used to support the comparisons in Chapter 5. It is intentionally selective: the main text keeps only the figures needed for the central empirical narrative, while the material below provides alternative normalizations, broader learner coverage, pairwise comparisons, horizon-scaling fits, and selected repeated-game profiles where these make the reported comparisons easier to verify. Unless stated otherwise, the experimental configuration is the same as in Chapter 5.

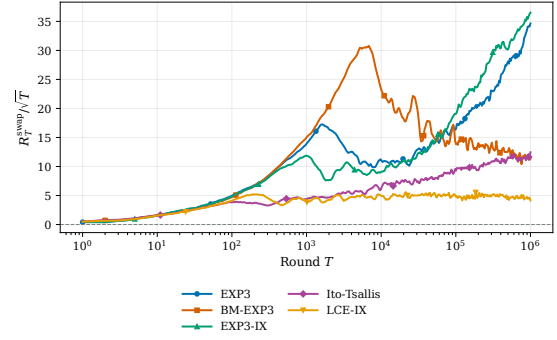
A.1 One-Player Results

A.1.1 Root-T Normalized Trajectories

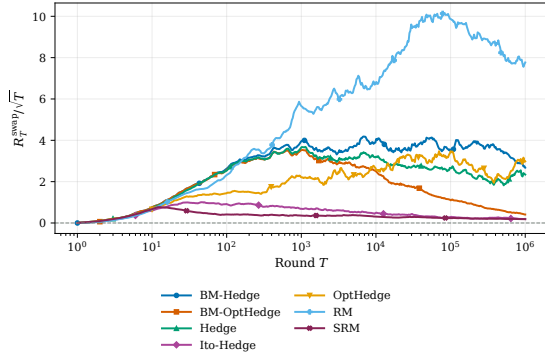
The $R_T/\sqrt{T}$ normalization is shown here for the aggregate swap-regret comparison and for the RM / SRM pairwise comparison.



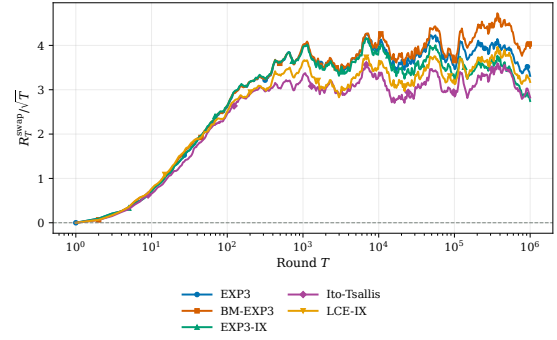
(a) HFA, full information



(b) HFA, bandit

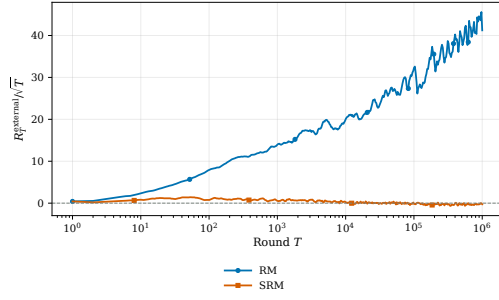


(c) LRW, full information

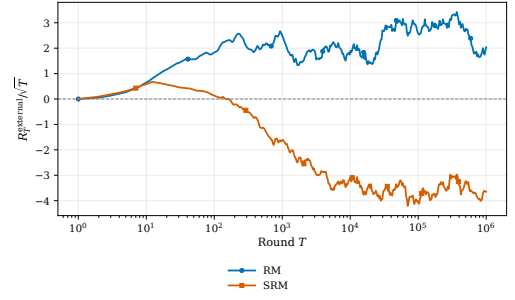


(d) LRW, bandit

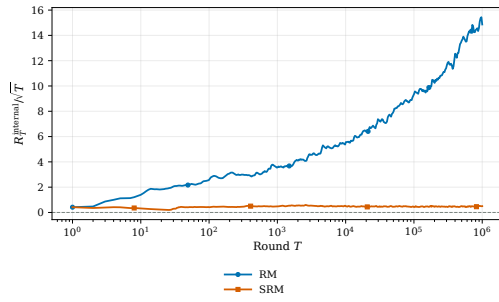
Figure A.1: Swap regret under $R_T/\sqrt{T}$ in both one-player environments.



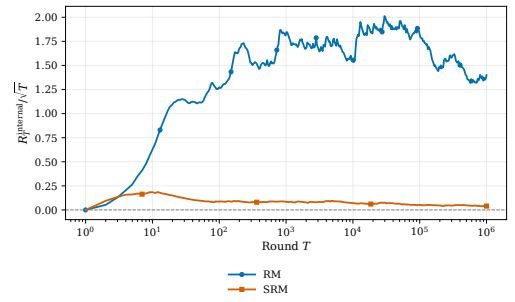
(a) HFA, external regret



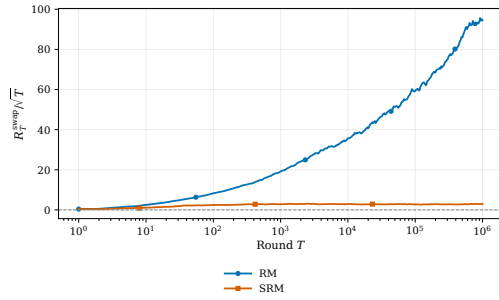
(b) LRW, external regret



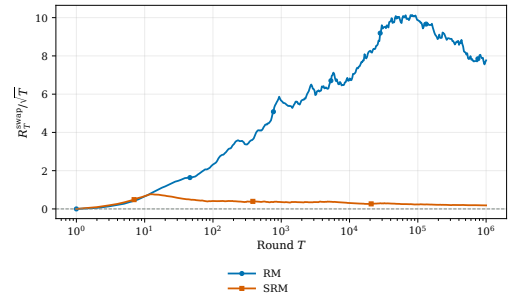
(c) HFA, internal regret



(d) LRW, internal regret



(e) HFA, swap regret

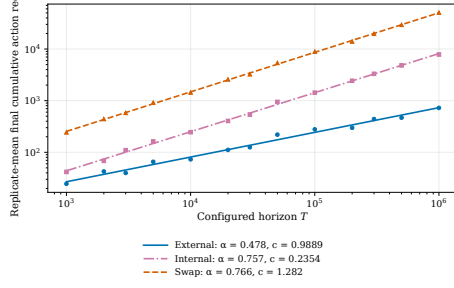


(f) LRW, swap regret

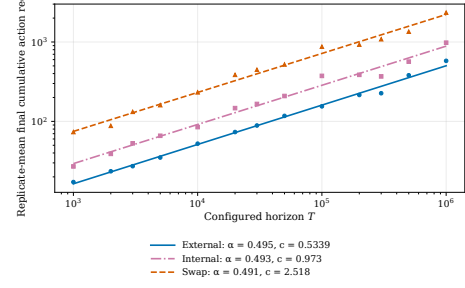
Figure A.2: RM and SRM under $R_T/\sqrt{T}$ in both one-player environments.

A.1.2 Individual Horizon-Scaling Fits

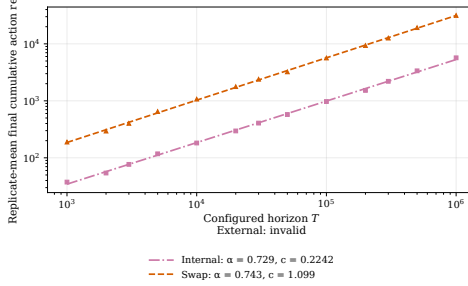
The following panels show the individual independent-horizon fits summarized by Table 5.1. Algorithms follow the same order as in the table, with HFA shown in the left column and LRW in the right column. The page breaks are used only for layout.



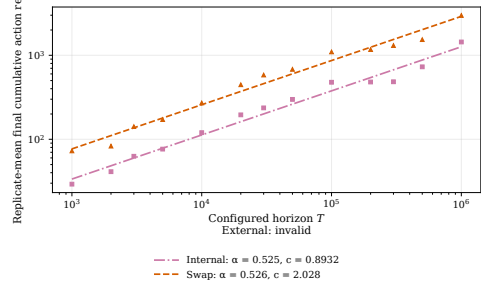
(a) HFA, Hedge



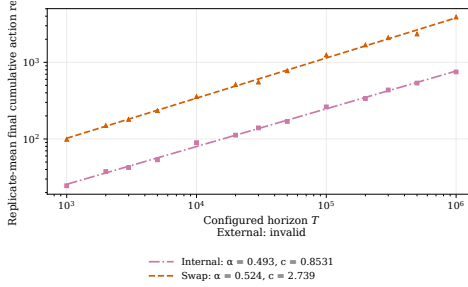
(b) LRW, Hedge



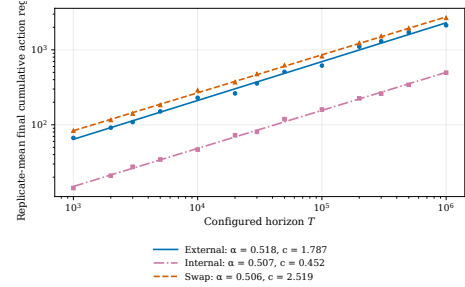
(c) HFA, OptHedge



(d) LRW, OptHedge

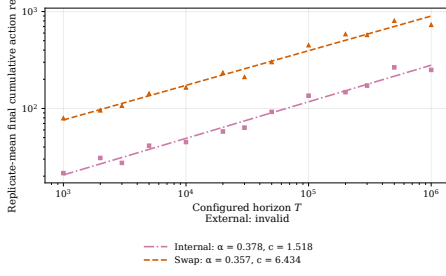


(e) HFA, BM-Hedge

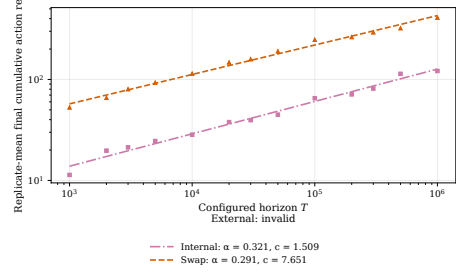


(f) LRW, BM-Hedge

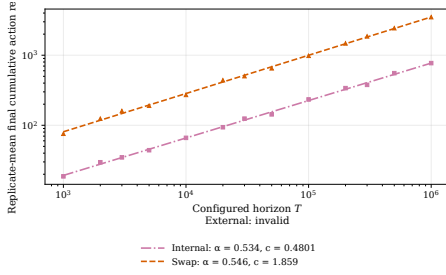
Figure A.3: Independent-horizon fits for the algorithms in Table 5.1.



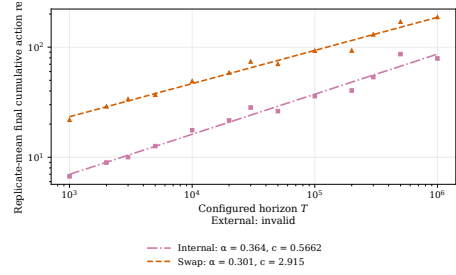
(g) HFA, BM-OptHedge



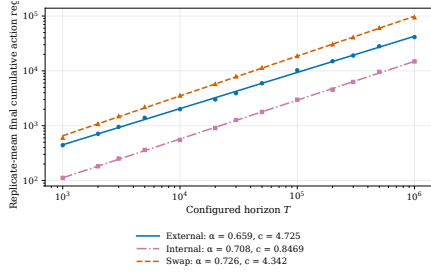
(h) LRW, BM-OptHedge



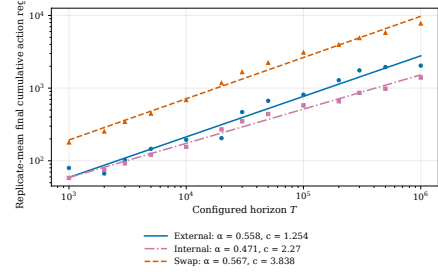
(i) HFA, Ito-Hedge



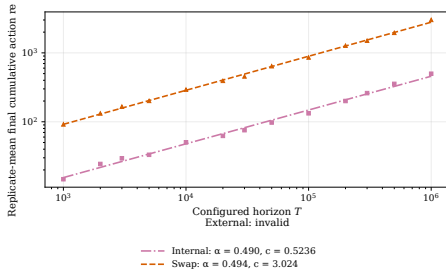
(j) LRW, Ito-Hedge



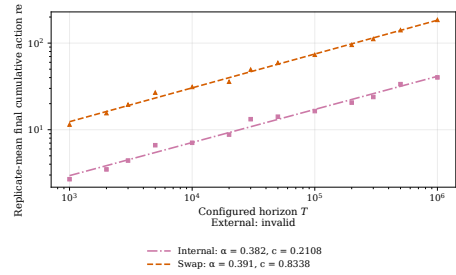
(k) HFA, RM



(l) LRW, RM

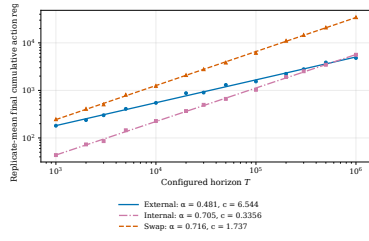


(m) HFA, SRM

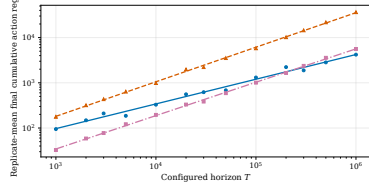


(n) LRW, SRM

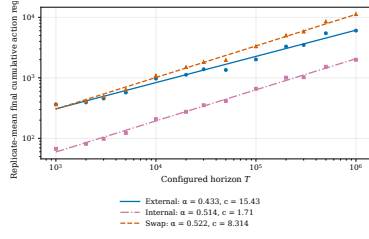
Figure A.3: Independent-horizon fits (continued).



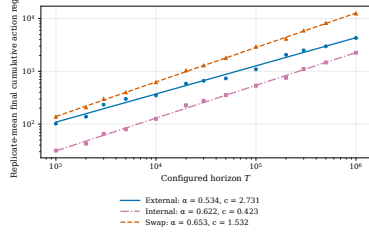
(o) HFA, EXP3



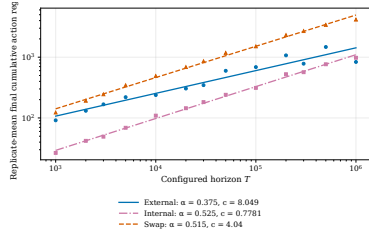
(q) HFA, EXP3-IX



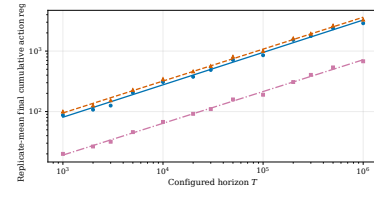
(s) HFA, BM-EXP3



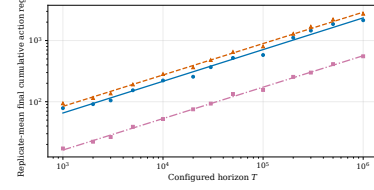
(u) HFA, Ito-Tsallis



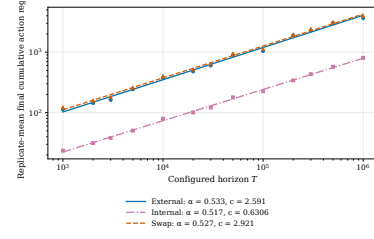
(w) HFA, LCE-IX



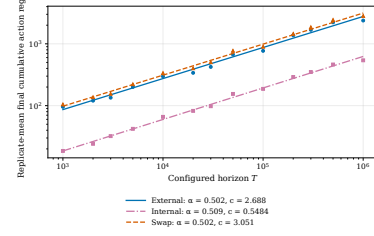
(p) LRW, EXP3



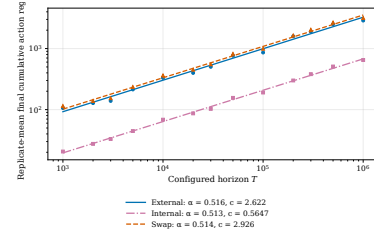
(r) LRW, EXP3-IX



(t) LRW, BM-EXP3



(v) LRW, Ito-Tsallis

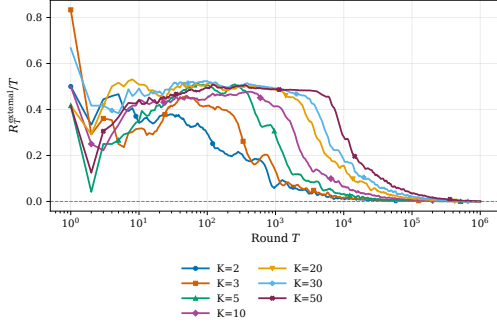


(x) LRW, LCE-IX

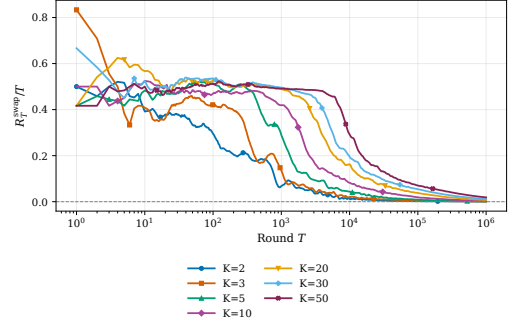
Figure A.3: Independent-horizon fits (continued).

A.1.3 Action-Space Scaling Examples

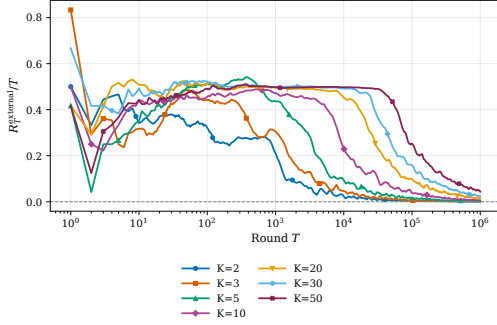
Figure A.4 provides selected additional examples for the action-space effect discussed in Section 5.1.4.



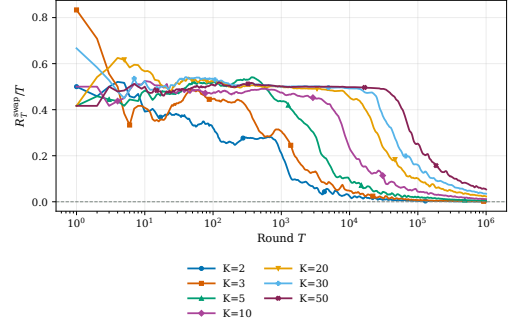
(a) BM-Hedge, external regret



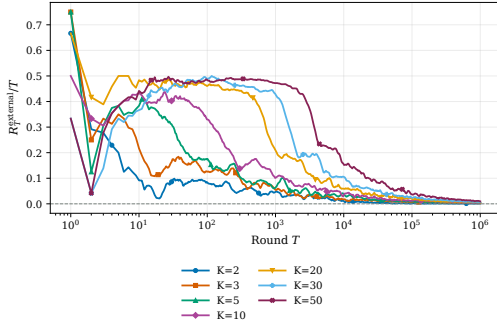
(b) BM-Hedge, swap regret



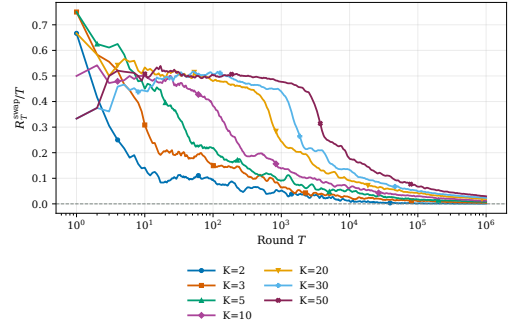
(c) BM-EXP3, external regret



(d) BM-EXP3, swap regret



(e) Ito-Tsallis, external regret



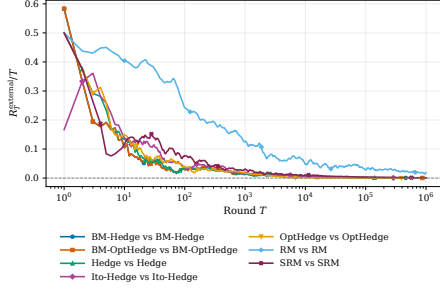
(f) Ito-Tsallis, swap regret

Figure A.4: Action-space scaling for BM-Hedge, BM-EXP3, and Ito-Tsallis in the historical-frequency environment.

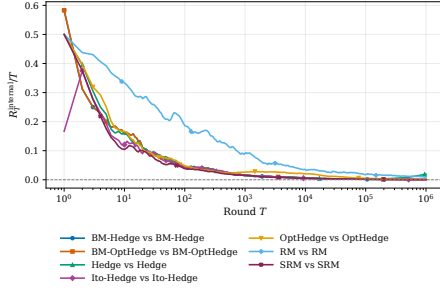
A.2 Repeated-Game Results

A.2.1 RPSLS Self-Play Regret Trajectories

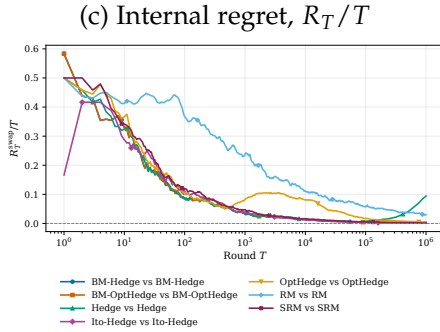
The panels supplement Table 5.2 with player 0 external-, internal-, and swap-regret trajectories for RPSLS self-play under both feedback models.



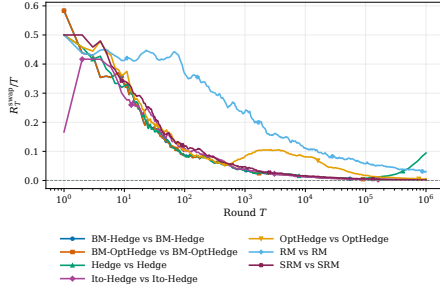
(a) External regret, R_T/T



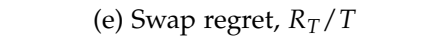
(b) External regret, $R_T/\sqrt{T}$



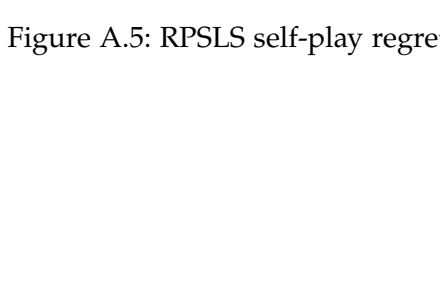
(c) Internal regret, R_T/T



(d) Internal regret, $R_T/\sqrt{T}$

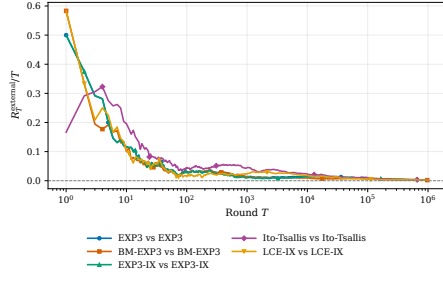


(e) Swap regret, R_T/T

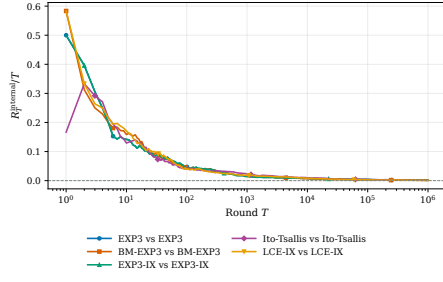


(f) Swap regret, $R_T/\sqrt{T}$

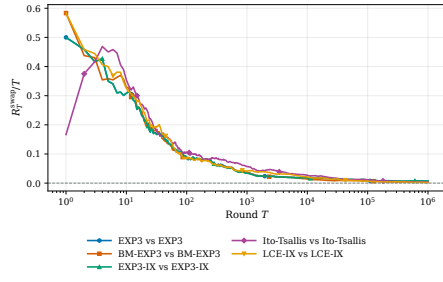
Figure A.5: RPSLS self-play regret for player 0 under full-information feedback at $T = 10^6$.



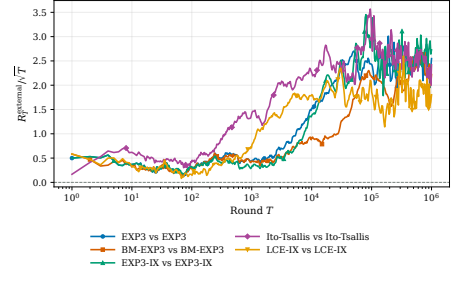
(a) External regret, R_T/T



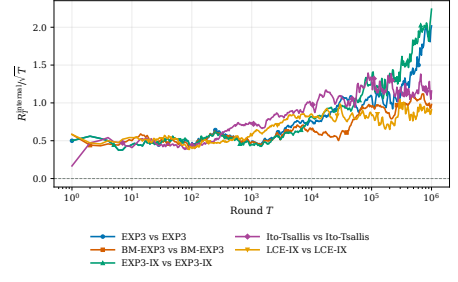
(c) Internal regret, R_T/T



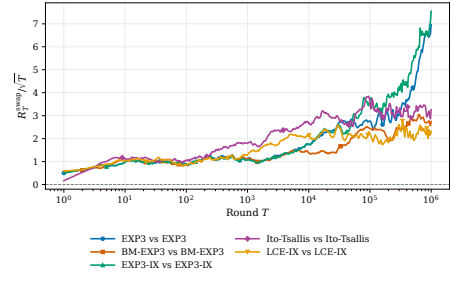
(e) Swap regret, R_T/T



(b) External regret, $R_T/\sqrt{T}$



(d) Internal regret, $R_T/\sqrt{T}$

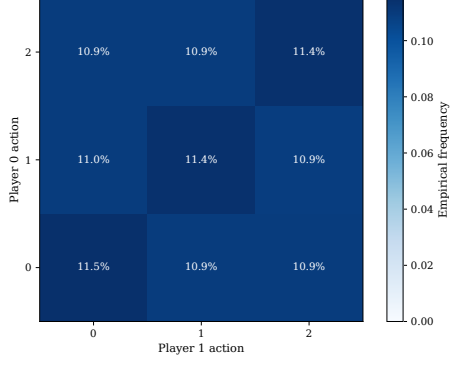


(f) Swap regret, $R_T/\sqrt{T}$

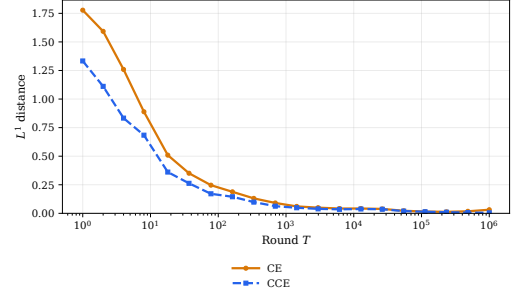
Figure A.6: RPSLS self-play regret for player 0 under bandit feedback at $T = 10^6$.

A.2.2 Selected RPS and RPSLS Equilibrium Profiles

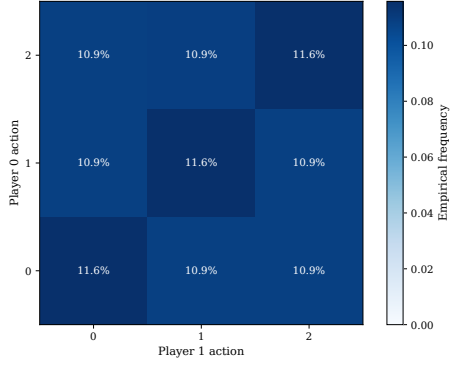
The panels below complement Section 5.2.2 with joint-action distributions and CE/CCE distance trajectories for EXP3 and EXP3-IX in RPS, and for Hedge in RPSLS.



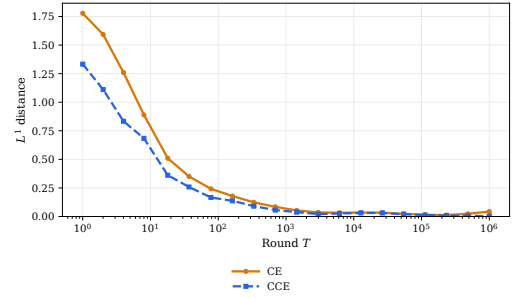
(a) RPS, EXP3, joint actions



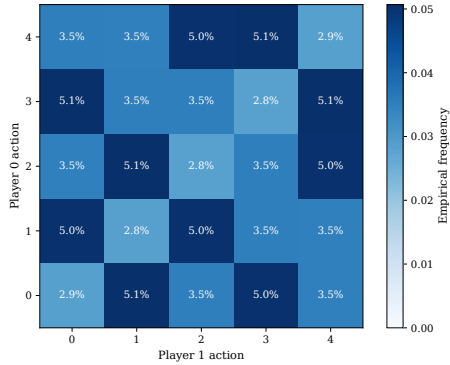
(b) RPS, EXP3, CE/CCE distance



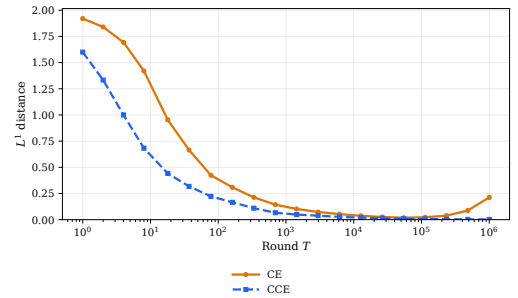
(c) RPS, EXP3-IX, joint actions



(d) RPS, EXP3-IX, CE/CCE distance



(e) RPSLS, Hedge, joint actions



(f) RPSLS, Hedge, CE/CCE distance

Figure A.7: Joint-action distributions and CE/CCE distances for EXP3 and EXP3-IX in RPS and Hedge in RPSLS self-play.

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